



Geopolitical risk and currency returns

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ABSTRACT

This study investigates the relationship between geopolitical risk (GPR) and currency excess returns. A zero-cost strategy that buys higher GPR currencies and sells lower GPR currencies generates a significant excess return of 5.72% per year. These returns contain information that goes beyond traditional currency investment strategies and cannot be explained by existing risk factors in asset pricing tests. Furthermore, the GPR factor is positively priced in broad cross sections of currency portfolios and in individual currencies. Further investigation reveals that the observed return predictability of GPR for currency returns stems from the country-specific idiosyncratic risk component and the regional risk component.

1. Introduction

The extant literature suggests that foreign exchange investors are compensated for taking time-varying risks (Lustig et al., 2011; Della Corte et al., 2016; Aloosh and Bekaert, 2022). However, few studies have considered geopolitical risk (GPR), which is a critical determinant influencing financial market dynamics, asset prices, and investment decisions (Berkman et al., 2011; Pástor and Veronesi, 2013).² Rising GPR may aggravate exchange rate fluctuations by influencing macroeconomic fundamentals, international trade and capital flows, and investor expectations (Calvo and Mendoza, 2000; Nitsch and Schumacher, 2004; Abadie and Gardeazabal, 2008). This influence results in large fluctuations in the expected returns on foreign exchange investments.³ Given recent geopolitical turmoil, such as the 9/11 attacks, the Paris terrorist attacks, and the 2017–2018 North Korea crisis (Caldara and Iacoviello, 2022), investors and researchers have placed a greater emphasis on GPR

(Bouoiyour et al., 2019; Baur and Smales, 2020).⁴ While most recent GPR research has focused on the stock (Narayan et al., 2022; Zaremba et al., 2022) and commodity markets (Brandt and Gao, 2019; Gkillas et al., 2020), there has been little research on the relationship between GPR and currency excess returns (CERs). We fill this void by investigating GPR's role in the foreign exchange (FX) market and demonstrating that GPR is a key driver and powerful predictor of cross-sectional CERs.

GPR, which is broadly defined as the risk associated with wars, terrorism, and tensions between states (Caldara and Iacoviello, 2022), differs from other measures of political instability and macroeconomic risk proposed in the existing literature in the following ways. First, GPR primarily focuses interstate issues, whereas political risk only addresses domestic political issues (Alsagr and Almazor, 2020). Second, GPR is based on sporadic but devastating events that might go undetected for decades (Guttentag and Herring, 1997). Thus, geopolitical events (e.g., terrorist acts) are more difficult to predict than other macroe-

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² Abbreviations: GPR for geopolitical risk and CERs for currency excess returns.

³ For example, following Russia's invasion of Crimea in March 2014, the Russian ruble depreciated by half against the US dollar within a year, resulting in losses for foreign exchange investors holding the Russian ruble.

⁴ According to the annual Systemic Risk Barometer Surveys published by the Depository Trust and Clearing Corporation, GPR has been a “top five” risk since 2016 and it surpassed cyber risk as the top threat to global financial stability in 2020. Similarly, in Gallup's (2017) survey, investors rank GPRs (e.g., military and diplomatic conflicts) as the top threat to the investing climate, ahead of the uncertainty of the domestic political climate and overall economic performance.

conomic events, making GPR evaluation difficult (Dissanayake et al., 2018). Third, the assessment of GPR has a greater negative impact than geopolitical acts themselves, most likely because the latter are perceived as resolving uncertainty (Dissanayake et al., 2018; Alsagr and Almazor, 2020). In this study, we build a GPR index for each country using a search-based newspaper coverage method. This index reflects both the risk of geopolitical events occurring and the new risk posed by their escalation.

To begin our investigation, we construct a monthly GPR index for each country by dividing the number of GPR-related news items retrieved from the Access World News database by the total number of news items. According to Caldara and Iacoviello (2022), keywords defining GPR-related news are classified into six categories: geopolitical threats, nuclear threats, war threats, terrorist threats, war acts, and terrorist acts. However, Caldara and Iacoviello's (2022) GPR index is constructed primarily from the perspective of North America and the United Kingdom, using 10 leading English-language newspapers. Access World News, alternatively, includes news from over 11,874 local, regional, national, and international sources. Because of the variety of sources, we can provide a more comprehensive and objective depiction of each country's geopolitical conditions from a broader perspective. Furthermore, we provide more country-specific GPR indices than Caldara and Iacoviello (2022).⁵

We divide currencies into five portfolios using the monthly GPR indices of 42 countries and regions from June 2002 to December 2019.⁶ The first (last) portfolio comprises currencies of countries within the lowest (highest) 20% of the GPR indices. Portfolio excess returns increase as GPR increases from low to high, indicating a positive relationship between GPR and cross-sectional CERs. The high-minus-low strategy of purchasing currencies from the last portfolio and selling those from the first yields an excess return of 5.72% per annum before considering the transaction costs and 4.73% after costs, which are economically large and statistically significant. We define the returns for this strategy as the GPR factor. This GPR-based strategy is distinct from the carry-trade strategy proposed by Lustig et al. (2011) or the good carry strategy proposed by Bekaert and Panayotov (2020). It incorporates new economic information that is not covered by traditional currency strategies and risk factors.

We next empirically test whether the GPR factor could explain the cross section of CERs in a standard asset pricing framework. We use a diverse set of 30 currency portfolios derived from the GPR, carry, short-term momentum, long-term momentum, value, and term spread strategies. The GPR factor is strongly priced across this broad cross section in its two-factor model involving the dollar factor, with a risk price of 0.90% per month (*t*-statistic of 3.45). This risk price suggests that an asset with a beta of 1 earns a risk premium of 10.80% per annum. Furthermore, even after controlling for a variety of existing risk factors in the literature, the pricing power of the GPR factor remains unaffected. These factors comprise the dollar and carry factors from Lustig et al. (2011), the global FX volatility from Menkhoff et al. (2012a), the TED spread used in Brunnermeier et al. (2008), the Bekaert–Engstrom–Xu US Risk Aversion Index from Bekaert et al. (2022) as well as two uncertainty indicators, including global political risk of Filippou et al. (2018) and global economic policy uncertainty of Baker et al. (2016). The findings thus far suggest a strong relationship between GPR and CERs.

⁵ By October 15, 2021, Dario Caldara and Matteo Iacoviello had provided country-specific GPR indices for only 19 emerging economies (<https://www.matteoiacoviello.com/gpr2019.htm>), a small subset compared to our dataset of 53 countries and regions worldwide. We also utilize their expanded set of GPR indices for 43 countries (by July 22, 2022, <https://www.matteoiacoviello.com/gpr.htm>) in our empirical tests in Section 6.1. Our main findings remain robust with this alternative measure.

⁶ We exclude from our sample the currencies of 11 countries that joined the Eurozone before June 2002.

Following the establishment of GPR's fundamental predictive properties, we delve deeper into the commonalities among GPR indices. We discover that GPR is affected by shocks that affect individual countries and those that affect the world or region as a whole by examining global and regional commonalities across country-specific GPR indices. As a result, we decompose the GPR indices into three parts: a global risk component, a regional risk component, and a country-specific idiosyncratic risk component. We find that the observed return predictability of GPR for currency returns stems from the country-specific idiosyncratic risk component and the regional risk component by constructing GPR portfolios based on each of these three risk components. Furthermore, even after removing the GPR commonalities, GPR's explanatory power remains robust. Finally, we perform robustness checks to confirm our main results.

This research contributes to two related fields of the literature. First, this study adds to the growing body of research on risk-based explanations for cross-sectional CERs. Most studies propose risk factors focusing on the carry-trade strategy, such as the carry factor in Lustig et al. (2011), unexpected global FX volatility in Menkhoff et al. (2012a), downside market risk in Lettau et al. (2014), and macroeconomic uncertainties in Berg and Mark (2018). In terms of the currency momentum strategy, Orlov (2016), Filippou et al. (2018), and Eriksen (2019) mention market illiquidity, global political risk, and cross-sectional return dispersion, respectively. A few studies have focused on more than one strategy, such as global imbalance in Della Corte et al. (2016) and business cycle risk in Colacito et al. (2020). This study adds to the existing literature by proposing a GPR factor that is significantly priced across a large currency cross section using multiple strategies, thereby providing new evidence of currency risk premium.

Second, this study adds to the body of knowledge on the role of GPR in financial markets. Existing research about GPR centers on the stock market (Balcilar et al., 2018; Narayan et al., 2022; Zaremba et al., 2022), gold and precious metals (Baur and Smales, 2020; Gkillas et al., 2020), crude oil (Bouoiyour et al., 2019; Brandt and Gao, 2019), the bond market (Chesney et al., 2011; Lee et al., 2021), mutual funds (Wang and Young, 2020), firm behavior (Shen et al., 2021), exchange rates (Salisu et al., 2022; Iyke et al., 2022), and cryptocurrencies (Aysan et al., 2019; Colon et al., 2021). However, the impact of GPR on CERs is still unknown. We fill this void by investigating GPR pricing ability in CER cross sections.

The remainder of this study is structured as follows. Section 2 discusses the construction of the GPR index, exchange rate data, and currency portfolios. Section 3 investigates the returns on GPR portfolios. Section 4 presents the asset pricing results. Section 5 explores the commonalities of GPR indices. Section 6 includes robustness tests. Section 7 concludes the paper. Lastly, the Internet Appendix provides additional results.

2. Data and currency portfolios

2.1. Data on GPR

According to Caldara and Iacoviello (2022), GPR is defined as the risk associated with wars, terrorism, and tensions among states that affect the normal course of international relations. GPR encompasses both the possibility of these events occurring and the new risks associated with an escalation of existing events. This definition includes geopolitical events frequently described as such, but excludes those only occasionally described as being geopolitical, such as climate change, global economic events, and major democratic political events.

To capture GPR, we adopt a method based on newspaper coverage frequency.⁷ We retrieve news items from Access World News, which has

⁷ The newspaper-based method has become a useful way to observe certain behaviors at a higher frequency than previously allowed (Da et al., 2015), as

Table 1
Search Groups.

Search Category	Search Terms
1. Geopolitical Threats	geopolitical AND (risk* OR concern* OR tension* OR uncertain*) “United States” AND tensions AND (military OR war OR geopolitical OR coup OR guerrilla OR warfare) AND (“Latin America” OR “Central America” OR “South America” OR Europe OR Africa OR “Middle East” OR “Far East” OR Asia)
2. Nuclear Threats	(“nuclear war” OR “atomic war” OR “nuclear conflict” OR “atomic conflict” OR “nuclear missile”) AND (fear* OR threat* OR risk* OR peril* OR menace*)
3. War Threats	“war risk*” OR “risk* of war” OR “fear of war” OR “war fear*” OR “military threat*” OR “war threat*” OR “threat of war” (“military action” OR “military operation” OR “military force”) AND (risk* OR threat*)
4. Terrorist Threats	“terrorist threat*” OR “threat of terrorism” OR “terrorism menace” OR “menace of terrorism” OR “terrorist risk” OR “terror risk” OR “risk of terrorism” OR “terror threat*”
5. War Acts	“beginning of the war” OR “outbreak of the war” OR “onset of the war” OR “escalation of the war” OR “start of the war” (war OR military) AND “air strike” (war OR battle) AND “heavy casualties”
6. Terrorist Acts	“terrorist act” OR “terrorist acts”

This table displays the search keywords used to define geopolitical risk news into six categories: geopolitical threats, nuclear threats, war threats, terrorist threats, war acts, and terrorist acts. As an example, in the first category of this table, we show a simple search term for the United States. Internet Appendix A contains the exact search queries for all sample countries. The asterisk (*) symbol denotes a wild-card character.

several advantages for building GPR indices. First, Access World News is the world’s largest and fastest-growing newspaper database, with over 11,874 local, regional, national, and international newspapers and news sources. As a result, it enables us to capture GPR more comprehensively and effectively worldwide. Second, Access World News gives us a more diverse perspective for developing GPR indices. In contrast to Caldara and Iacoviello’s (2022) GPR index, which is based on text searches conducted on 10 major newspapers primarily from North America and the United Kingdom, our GPR index is built with both local and global news sources, providing a more objective reflection of GPR in each country.⁸ Third, Access World News has a strong timeliness, closely tracking current events from 1980 to the present. As a result, it meets our requirements for building GPR indices at both the time-series and cross-sectional levels.

Specifically, for each country, we search Access World News for the number of GPR-related news items and all news each month. An item must mention both the country of interest and at least one keyword that defines GPR-related news to be considered GPR-related news for that country. These keywords, based on those used by Caldara and Iacoviello (2022), are organized into six categories: geopolitical threats, nuclear threats, war threats, terrorist threats, war acts, and terrorist acts, as shown in Table 1. We also exclude from our search terms that are frequently associated with false positives, such as “civil war” and “human rights,” as well as words like “movie,” “museum,” “anniversary,” “memorial,” and “art.” Internet Appendix A contains the search queries for all sample countries. The total number of news items used to scale the raw GPR-related news counts for each country only requires

mentioning the country of interest. We perform all searches in *English*, as Access World News translates articles written in foreign languages into English. We confine the search scope for keywords appearing in the news to *Lead/First Paragraph*. Additionally, following prior literature, we limit our searches to the *newspaper* category. We thus construct the GPR index for each country j in month t as the following:

$$gpr_{j,t} = \frac{\text{number of geopolitical risk related news items}_{j,t}}{\text{total number of news items}_{j,t}}. \quad (1)$$

That is, the GPR index reflects the proportion of news discussing rising GPRs in the total amount of news published each month.

We validate the computer-generated GPR index further by conducting an extensive human audit of newspaper articles that are likely to discuss GPRs, thereby reinforcing the validity of our article selection process. Because we build the GPR index on a country-by-country basis, we also perform the human audit on a country-by-country basis, using Australia as an example. Using our search criteria, we searched Access World News and retrieved 2,120 GPR-related news articles for Australia from 1985 to 2019. We then selected 200 news articles randomly from the set chosen by the automated text-search algorithm and classified them as either discussing high or rising geopolitical tensions. We adhere to Caldara and Iacoviello’s (2022) audit guidelines, which are as follows: i) If the article discusses or highlights recent past, current, or expected future geopolitical threats, nuclear threats, war threats, terrorist threats, war acts, or terrorist acts, label it as 1. ii) If the article discusses falling tensions, label it as -1 . iii) If the article does not mention GPRs or is uninformative about whether they are rising or falling, mark it as 0. Following this audit, 71% of the articles that comprise Australia’s computer-generated GPR-related news articles mention high or rising GPRs. Only 2% of the articles mention low or decreasing geopolitical tensions, while other false positives fall into various categories, such as discussions of past geopolitical events and related personal experiences that have no immediate connection to current developments. The low frequency of articles discussing positive geopolitical developments supports our claim that our choice of words captures negative risks to the geopolitical outlook, because positive events are discussed using a different set of words (e.g., “negotiations” or “peace talks”).

We construct monthly GPR indices for 53 countries and regions from January 1985 to December 2019. The Eurozone GPR index is calculated

well as to measure the amount of attention that newspaper editors, reporters, and readers pay to a particular topic (Fang et al., 2014; Kaniel and Parham, 2017). The newspaper-based method has been used to create many uncertainty and risk indices, including the global and country-specific economic policy uncertainty indices developed by Baker et al. (2016), the US equity market volatility index developed by Baker et al. (2019), and the US monetary policy uncertainty index proposed by Husted et al. (2018).

⁸ We construct the GPR index using news sources from Access World News in July 2020, which total 11,874, with 641 from Africa, 916 from Asia, 745 from Australia/Oceania, 1,218 from Europe, 428 from the Middle East, 7,876 from North America, and 50 from South America.

Table 2
Descriptive Statistics: Geopolitical Risk Indices.

Country	Time period	Mean	Std	Min	Max	Median
Argentina	1985.01–2019.12	0.11	0.36	0.00	4.69	0.00
Australia	1985.01–2019.12	0.07	0.12	0.00	1.23	0.03
Austria	1985.01–2019.12	0.12	0.28	0.00	2.70	0.00
Belgium	1985.01–2019.12	0.19	0.51	0.00	5.54	0.00
Brazil	1985.01–2019.12	0.07	0.17	0.00	2.07	0.00
Canada	1985.01–2019.12	0.08	0.13	0.00	1.38	0.04
Chile	1985.01–2019.12	0.08	0.28	0.00	3.50	0.00
Chinese mainland	1985.01–2019.12	0.35	0.36	0.00	3.78	0.27
China's Hong Kong	1985.01–2019.12	0.06	0.12	0.00	0.92	0.00
Colombia	1985.01–2019.12	0.19	0.37	0.00	2.71	0.00
Croatia	1985.01–2019.12	0.19	1.03	0.00	19.05	0.00
Cyprus	1985.01–2019.12	0.30	1.11	0.00	17.17	0.00
Czech Republic	1985.01–2019.12	0.10	0.33	0.00	3.70	0.00
Denmark	1985.01–2019.12	0.09	0.29	0.00	3.52	0.00
Egypt	1985.01–2019.12	0.40	0.55	0.00	4.30	0.26
Estonia	1985.01–2019.12	0.25	1.16	0.00	12.20	0.00
the Eurozone	1999.01–2019.12	0.17	0.19	0.01	2.00	0.13
Finland	1985.01–2019.12	0.09	0.38	0.00	4.35	0.00
France	1985.01–2019.12	0.21	0.29	0.00	2.73	0.13
Germany	1985.01–2019.12	0.23	0.31	0.00	2.95	0.15
Greece	1985.01–2019.12	0.29	0.73	0.00	7.32	0.00
Hungary	1985.01–2019.12	0.16	0.42	0.00	4.03	0.00
Iceland	1985.01–2019.12	0.12	0.70	0.00	8.89	0.00
India	1985.01–2019.12	0.37	0.78	0.00	10.17	0.19
Indonesia	1985.01–2019.12	0.24	0.53	0.00	4.34	0.00
Ireland	1985.01–2019.12	0.07	0.18	0.00	3.07	0.02
Israel	1985.01–2019.12	0.42	0.38	0.00	2.72	0.30
Italy	1985.01–2019.12	0.10	0.19	0.00	2.50	0.05
Japan	1985.01–2019.12	0.27	0.36	0.00	4.27	0.17
Kuwait	1985.01–2019.12	0.84	1.71	0.00	16.11	0.32
Latvia	1985.01–2019.12	0.21	0.82	0.00	9.09	0.00
Lithuania	1985.01–2019.12	0.23	1.06	0.00	15.25	0.00
Mexico	1985.01–2019.12	0.06	0.10	0.00	0.90	0.03
Morocco	1985.01–2019.12	0.28	1.13	0.00	12.21	0.00
Netherlands	1985.01–2019.12	0.16	0.58	0.00	9.42	0.00
New Zealand	1985.01–2019.12	0.04	0.14	0.00	1.94	0.00
Norway	1985.01–2019.12	0.07	0.21	0.00	1.94	0.00
Pakistan	1985.01–2019.12	0.74	1.19	0.00	12.30	0.42
Peru	1985.01–2019.12	0.07	0.29	0.00	3.80	0.00
the Philippines	1985.01–2019.12	0.28	0.58	0.00	8.65	0.10
Portugal	1985.01–2019.12	0.10	0.38	0.00	3.92	0.00
Singapore	1985.01–2019.12	0.13	0.25	0.00	2.50	0.05
Slovakia	1985.01–2019.12	0.05	0.22	0.00	1.96	0.00
Slovenia	1985.01–2019.12	0.15	0.70	0.00	7.89	0.00
South Africa	1985.01–2019.12	0.08	0.13	0.00	0.97	0.03
South Korea	1985.01–2019.12	0.21	0.29	0.00	2.40	0.12
Spain	1985.01–2019.12	0.10	0.26	0.00	2.96	0.03
Sweden	1985.01–2019.12	0.06	0.18	0.00	1.54	0.00
Switzerland	1985.01–2019.12	0.09	0.23	0.00	1.94	0.00
Thailand	1985.01–2019.12	0.12	0.28	0.00	3.20	0.00
Turkey	1985.01–2019.12	0.30	0.50	0.00	5.25	0.17
the United Kingdom	1985.01–2019.12	0.14	0.19	0.00	1.73	0.09
the United States	1985.01–2019.12	0.15	0.12	0.01	0.98	0.11

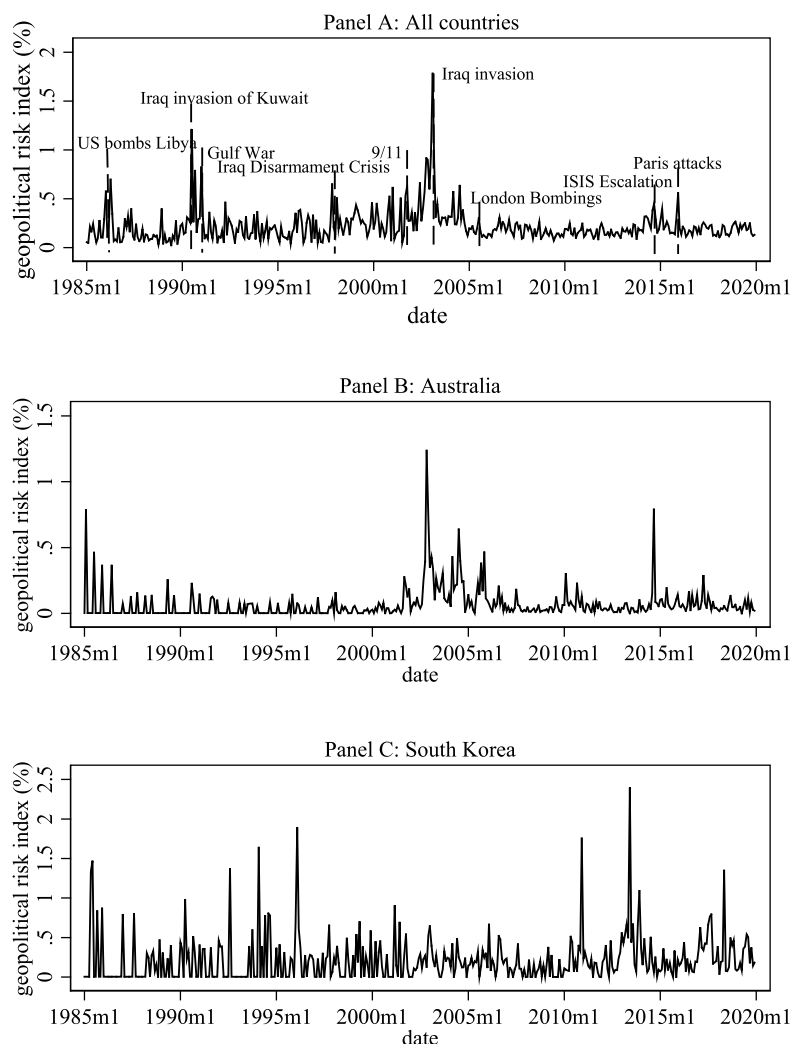
The descriptive statistics of the monthly geopolitical risk (GPR) indices for 53 countries and regions around the world are presented in this table. For each GPR index, we provide the time period, as well as the mean (*Mean*), standard deviation (*Std*), minimum (*Min*), maximum (*Max*), and median (*Median*) values of GPR indices. GPR indices are expressed in percentage per month.

by averaging the GDP-weighted GPR indices of the countries that join the Eurozone. The World Bank database obtains data on gross domestic product (GDP) for Eurozone countries. The descriptive statistics are shown in Table 2. Our GPR index construction method and the keywords used to identify GPR-related news are based on Caldara and Iacoviello (2022); thus, we verify our GPR index by examining the correlation between our GPR index and theirs. We find that the Recent GPR index developed by Dario Caldara and Matteo Iacoviello is highly correlated with the GDP-weighted average of our country-specific GPR indices at 0.78 and the equal-weighted average at 0.58. This suggests

that our GPR index performs as well as they do in capturing geopolitical risk.⁹

In addition, we show the time series of the GPR indices in Fig. 1. We begin by taking a broad look at the GPR index by calculating the equally weighted average of the GPR indices of all available countries for each month. As shown in Panel A, the GPR index fluctuates over time and

⁹ Data on the Recent GPR index developed by Dario Caldara and Matteo Iacoviello were downloaded from <https://www.matteoiacoviello.com/gpr2019.htm> on July 22, 2022.



The monthly geopolitical risk (GPR) indices are plotted in this graph from January 1985 to December 2019. Panel A depicts the GPR index, which is calculated each month by taking the equally weighted average of the GPR indices of all available countries. Panel B depicts the Australian GPR index, while Panel C depicts the South Korean GPR index. The GPR indices are expressed as percentage points.

Fig. 1. Geopolitical Risk Indices.

peaks around major geopolitical events such as the US air attack on Libya in April 1986, the Iraq war in 2003, and the terrorist attack in Paris in 2015; this observation indicates that the GPR index captures GPR globally. We also plot each country's GPR index time series, using Australia and South Korea as examples in Panels B and C. We find that Australia, as a major participant in the Iraq war in 2003, experienced a peak in its GPR index during this period. Meanwhile, South Korea was not involved in this war and thus had a fairly stable GPR index. This evidence demonstrates that, when compared to a global GPR index, country-specific GPR indices enable investors to capture each country's unique risk profile more effectively.

2.2. Data on exchange rates

We obtain daily bid, mid, and ask quotes on spot and one-month forward exchange rates against the US dollar from WM/Reuters, which we access through Datastream.¹⁰ We define exchange rates as units of

foreign currency per US dollar, so an increase in the exchange rate indicates that the home currency has appreciated (i.e., the US dollar). Bid, mid, and ask quotes account for transaction costs. The empirical analysis employs monthly data derived from end-of-month rate sampling.

For empirical tests, we process the raw exchange rate data in the manner described by Lustig et al. (2011) and subsequent research. First, we remove Eurozone countries from the dataset once they adopt the euro. As a result, the sample used for empirical analysis from June 2002 to December 2019 includes 42 currencies, including those of Argentina, Australia, Brazil, Canada, Chile, the Chinese mainland, China's Hong Kong, Colombia, Croatia, Cyprus, Czech Republic, Denmark, Egypt, Estonia, the Eurozone, Hungary, Iceland, India, Indonesia, Israel, Japan, Kuwait, Latvia, Lithuania, Mexico, Morocco, New Zealand, Norway, Pakistan, Peru, the Philippines, Singapore, Slovakia, Slovenia, South Africa, South Korea, Sweden, Switzerland, Thailand, Turkey, the United Kingdom, and the United States.¹¹ Table 3 presents the individual start

¹⁰ We rely on WM/Reuters because it covers a broader range of countries and meets our cross-sectional data needs, and we supplement WM/Reuters with Barclays Bank International data obtained through Datastream.

¹¹ We have extended the GPR index data to October 1983; exchange rate data have been available since then. However, as illustrated in Fig. IA1 (see Internet Appendix), the average number of countries with nonmissing and nonzero GPR

Table 3
FX Spot and Forward Rates: Data Sources.

Country	Currency Code	Spot	1M Forward	Start Date	End Date
Argentina	ARS	ARGPES\$	USARS1F	31/03/2004	31/12/2019
Australia	AUD	AUSTDOI	USAUD1F	28/06/2002	31/12/2019
Brazil	BRL	BRACRU\$	USBRL1F	31/03/2004	31/12/2019
Canada	CAD	CNDOLL\$	USCAD1F	28/06/2002	31/12/2019
Chile	CLP	CHILPE\$	USCLP1F	31/03/2004	31/12/2019
Chinese mainland	CNY	CHIYUA\$	USCNY1F	28/06/2002	31/12/2019
China's Hong Kong	HKD	HKDOLL\$	USHKD1F	28/06/2002	31/12/2019
Colombia	COP	COLUPE\$	USCOP1F	31/03/2004	31/12/2019
Croatia	HRK	CROATK\$	USHRK1F	31/03/2004	31/12/2019
Cyprus	CYP	CYPRUS\$	USCYP1F	31/03/2004	31/12/2007
Czech Republic	CZK	CZECHC\$	USCZK1F	28/06/2002	31/12/2019
Denmark	DKK	DANISH\$	USDKK1F	28/06/2002	31/12/2019
Egypt	EGP	EGYPTN\$	USEGP1F	31/03/2004	31/12/2019
Estonia	EEK	ESTOKR\$	USEEK1F	31/03/2004	31/12/2010
the Eurozone	EUR	EUDOLLR	EUDOL1F	28/06/2002	31/12/2019
Hungary	HUF	HUNFOR\$	USHUF1F	28/06/2002	31/12/2019
Iceland	ISK	ICEKRO\$	USISK1F	31/03/2004	31/12/2019
India	INR	INDRUP\$	USINR1F	28/06/2002	31/12/2019
Indonesia	IDR	INDORU\$	USIDR1F	28/06/2002	31/12/2019
Israel	ILS	ISRSHE\$	USILS1F	31/03/2004	31/12/2019
Japan	JPY	JAPAYE\$	USJPY1F	28/06/2002	31/12/2019
Kuwait	KWD	KUWADI\$	USKWD1F	28/06/2002	31/12/2019
Latvia	LVL	LATVLA\$	USLVL1F	31/03/2004	31/12/2013
Lithuania	LTL	LITITA\$	USLTL1F	31/03/2004	31/12/2014
Mexico	MXN	MEXPES\$	USMXN1F	28/06/2002	31/12/2019
Morocco	MAD	MOROCD\$	USMAD1F	31/03/2004	31/12/2019
New Zealand	NZD	NZDOLLI	USNZD1F	28/06/2002	31/12/2019
Norway	NOK	NORKRO\$	USNOK1F	28/06/2002	31/12/2019
Pakistan	PKR	PAKRUP\$	USPKR1F	31/03/2004	31/12/2019
Peru	PEN	PERUSO\$	USPEN1F	31/03/2004	31/12/2019
the Philippines	PHP	PHILPE\$	USPHP1F	28/06/2002	31/12/2019
Singapore	SGD	SINGDO\$	USSGD1F	28/06/2002	31/12/2019
Slovakia	SKK	SLOVKO\$	USSKK1F	31/03/2004	31/12/2008
Slovenia	SIT	SLOVTO\$	USSIT1F	31/03/2004	31/12/2006
South Africa	ZAR	COMRAN\$	USZAR1F	28/06/2002	31/12/2019
South Korea	KRW	KORSWO\$	USKRW1F	28/06/2002	31/12/2019
Sweden	SEK	SWEKRO\$	USSEK1F	28/06/2002	31/12/2019
Switzerland	CHF	SWISSF\$	USCHF1F	28/06/2002	31/12/2019
Thailand	THB	THABAH\$	USTHB1F	28/06/2002	31/12/2019
Turkey	TRY	TURKLI\$	USTRY1F	28/06/2002	31/12/2019
the United Kingdom	GBP	UKDOLLR	UKUSD1F	28/06/2002	31/12/2019

This table displays the currency code, Datastream mnemonics for spot and forward exchange rate data, and the start and end dates of exchange rate data for each country. Daily spot and forward exchange rates are obtained from WM/Reuters using Datastream.

and end dates for each currency as well as Datastream mnemonics. There are no descriptions for the US dollar because the exchange rate data are relative to it. Second, we eliminate observations when the covered interest rate parity (CIP) condition is violated severely.¹² Third, throughout the sample, some of the sample currencies pegged their exchange rates partially or entirely to the US dollar. We kept them in our sample because forward contracts were easily accessible to investors (Lustig et al., 2011; Filippou and Taylor, 2023).

We denote the log of the spot and 1-month forward exchange rates in units of foreign currency per US dollar at time t as s_t and f_t , respectively. Following Lustig et al. (2011), the log excess return rx on buying a foreign currency in the forward market and then selling it in

the spot market after one month is simply as $rx_{t+1} = f_t - s_{t+1}$. To satisfy the Euler equation, which is for levels of returns rather than logs, we use discrete returns (rather than log returns, as stated above) in all of our empirical analyses below (Lustig et al., 2011; Menkhoff et al., 2012a). The discrete excess return RX_{t+1} is computed as (Menkhoff et al., 2012a; Ahmed and Valente, 2015; Filippou and Taylor, 2017, 2023)

$$RX_{t+1} = \frac{F_t - S_{t+1}}{S_t}, \quad (2)$$

where S_t and F_t are the level of spot and forward exchange rates, respectively. This excess return can also be stated as the difference between the forward discount and the change in the spot rate:

$$RX_{t+1} = \frac{F_t - S_t}{S_t} - \frac{S_{t+1} - S_t}{S_t}. \quad (3)$$

According to the CIP condition, the forward discount is approximately equal to the difference between domestic and foreign interest rates; that is, $(F_t - S_t)/S_t \simeq i_t^* - i_t$, where i_t^* and i_t , respectively, represent the foreign and domestic risk-free rates, over the maturity of the forward contract. Given that CIP is held on a daily and lower frequency basis (Akram et al., 2008), the excess returns could be written as $RX_{t+1} \simeq (i_t^* - i_t) - (S_{t+1} - S_t)/S_t$.

We also compute CERs adjusted for transaction costs using the spot and 1-month forward exchange rates' bid, mid, and ask quotes

index values from October 1983 to May 2002 is 6.91, which is significantly less than the average from June 2002 to December 2019, which is 21.51. To ensure an adequate sample size for portfolio construction, we begin the empirical analysis in June 2002. Furthermore, using the entire sample of 53 countries and regions from October 1983 to December 2019, we confirm in Table IA1 (see Internet Appendix) the robustness of our main results.

¹² The following observations are excluded from the sample: Argentina from September 2008 to December 2008 and from May 2012 to June 2014; Egypt from October 2016 to March 2017; Indonesia from June 2002 to September 2002 and from November 2008 to February 2009; South Africa from June 2002 to May 2005.

Table 4
FX Investment Strategies: Data Sources.

Strategy	Signal	Source
Geopolitical Risk Strategy	$gpr_{j,t}$	Authors' calculation based on the raw news volume data crawled from Access World News. It is computed as the proportion of news discussing geopolitical risks in the total news each month t for each country j .
Carry Trade	$(F_t - S_t) / S_t$	Authors' calculation based on spot and forward exchange rate quotes from WM/Reuters accessed via Datastream.
Short-term Momentum	$RX_{t-3:t}$	Authors' calculation based on spot and forward exchange rate quotes from WM/Reuters accessed via Datastream.
Long-term Momentum	$RX_{t-12:t}$	Authors' calculation based on spot and forward exchange rate quotes from WM/Reuters accessed via Datastream.
Currency Value	$-(\log Q_t - \log Q_{t-5yr})$	To calculate real exchange rates for the value strategy, we use Consumer Price Index data from the IMF's International Financial Statistics. The real exchange rates, denoted by Q_t , is calculated as $Q_t = P_t^* / (S_{t,t} P_t)$, where P_t^* and P_t are the price levels of consumer goods in foreign currency and price level of consumer goods in the US stated in United States Dollars, respectively.
Term Spread	$-(i_{10yr} - i_{3mo})$	We use Global Financial Data's 3-month government bill rates and 10-year government bond rates (or 5-year government bond rates when 10-year government bond rates are unavailable).

This table provides information about the data used for the construction of the six currency investment strategies analyzed in the study. The first column (*Strategy*) reports the currency strategy's name. The second column (*Signal*) specifies the trading signal used by the strategy, while the third column (*Source*) reports the sources of the data used for the construction of the signals.

(Menkhoff et al., 2012a; Della Corte et al., 2016). In particular, buying a foreign currency forward at time t at bid price (F_t^b) and offsetting the position at time $t+1$ in the spot market at ask price (S_{t+1}^a) renders a net excess return of the form $RX_{t+1}^l = (F_t^b - S_{t+1}^a) / S_t^b$, where l denotes a long position in the foreign currency. If the investor buys a foreign currency at time t but maintains the position at time $t+1$, the net excess return is calculated as $RX_{t+1}^l = (F_t^b - S_{t+1}^b) / S_t^b$. Similarly, if the investor closes the position in foreign currency at time $t+1$, which already exists at time t , the net excess return is computed as $RX_{t+1}^l = (F_t - S_{t+1}^a) / S_t^b$. Besides, selling a foreign currency forward at time t at ask price (F_t^a) and offsetting the position at time $t+1$ in the spot market at bid price (S_{t+1}^b) renders a net excess return of the form $RX_{t+1}^s = -(F_t^a - S_{t+1}^b) / S_t^a$, where s denotes a short position in the foreign currency. If the foreign currency leaves the strategy at time t , and the short position is rolled over at time $t+1$, the net excess return is defined as $RX_{t+1}^s = -(F_t^a - S_{t+1}^b) / S_t^a$. Similarly, if the investor closes a short position in foreign currency at time $t+1$, which already exists at time t , the net excess return is computed as $RX_{t+1}^s = -(F_t - S_{t+1}^b) / S_t^a$. We assume that investors open new positions in each currency in the first month and close them in the last month. The total number of currencies in our portfolios fluctuates over time, and we only include currencies with bid and ask quotes on the spot and forward exchange rates from the current and subsequent periods. Portfolio 1 currencies are adjusted for transaction costs in short positions, whereas currencies in the other portfolios are adjusted for transaction costs in long positions.

2.3. Currency portfolios

This section describes the development of six currency strategies, which comprise a broad currency cross section of 30 portfolios used as test assets in our empirical analysis. All of these portfolios are rebalanced monthly, spanning from June 2002 to December 2019.

We first construct five GPR portfolios by sorting currencies according to their GPR indices. At the end of each month t , currencies from the countries within the top 20% of the GPR indices are allocated to Portfolio 5, whereas those within the bottom 20% of the GPR indices are allocated to Portfolio 1. The excess returns for each portfolio are calculated as the equal-weighted average of all individual CERs within that portfolio. Furthermore, we develop the GPR factor through the implementation of a zero-cost strategy involving a long position in Portfolio 5 and a short position in Portfolio 1.

Similarly, we build another 25 currency portfolios by employing five widely used currency investment strategies: carry trade (Lustig et al., 2011; Menkhoff et al., 2012a), short-term and long-term momentum (Asness et al., 2013; Menkhoff et al., 2012b), currency value (Asness et al., 2013; Menkhoff et al., 2017), and term spread (Bekaert et al., 2007;

Lustig et al., 2019). The signals used to allocate currencies into portfolios differ between these strategies, but the sorting schemes are similar to the construction of GPR portfolios. Specifically, at time t , currencies are allocated to five portfolios based on the signal for the selected strategy. The excess returns for a generic portfolio are then computed as the equally weighted average of the individual CERs allocated to that portfolio. Table 4 shows the trading signals and data sources used to develop each strategy. By design, the portfolios should contain currencies with increasing riskiness as we progress from Portfolio 1 to Portfolio 5.

3. GPR and currency returns

3.1. Portfolio excess returns and the GPR factor

Table 5 presents the descriptive statistics on the five GPR portfolios and the GPR factor. Panel A reports portfolio returns before taking into account transaction costs, whereas Panel B reports returns net of transaction costs using bid and ask rates. The average excess returns of the five portfolios tend to increase when moving from Portfolio 1 (0.79% and 1.26% per annum, respectively) to Portfolio 5 (6.50% and 5.99% per annum, respectively) for Panels A and B. Hence, the zero-cost strategy that is long Portfolio 5 and short Portfolio 1 yields an annual excess return of 5.72% before transaction costs and 4.73% after costs, which are both economically large and statistically significant. We denote the returns for this long-short strategy as the GPR factor. Moreover, the annualized Sharpe ratios are sizable for Panels A and B at 75% and 62%, respectively. Overall, higher GPR currencies generate greater excess returns than lower GPR currencies. We divide CERs into two categories: spot exchange rate returns and forward discounts (see rows *spot* and *fd*, respectively in Table 5). The forward discount component exhibits a monotonic pattern across all five portfolios, similar to carry-trade portfolios (Lustig et al., 2011). However, the currency compositions of GPR portfolios and carry-trade portfolios in Table IA2 (see Internet Appendix) show that the currencies of the highest GPR economies are not always the currencies with the highest interest rates. The spot exchange rate component shows no discernible trend from Portfolio 1 to Portfolio 5, indicating a distinction in the features of currency momentum portfolios (Menkhoff et al., 2012b).

We also use the formula for calculating CERs when exchange rates are defined as units of US dollars per unit of foreign currency (Bekaert and Hodrick, 1993) as $RX_{t+1} = (S_{t+1} - F_t) / S_t$, where F_t and S_t are the level of the forward and spot exchange rate in units of US dollars per foreign currency, respectively. The results in Table 6 show the same return characteristics as our baseline results in Table 5, indicating that our baseline results are still robust with this CER formula. Furthermore, our baseline results are robust when i) using interest rate differentials to calculate CERs (Table IA3 in the Internet Appendix) and ii) constructing GPR portfolios by sorting currencies at the end of each 3-month or 12-

Table 5
Descriptive Statistics: Geopolitical Risk Portfolios.

	P1	P2	P3	P4	P5	Avg.	GPR
Panel A: without Transaction Costs							
mean (%)	0.79	2.74	3.44**	3.11	6.50***	3.32*	5.72**
<i>t</i> -stat	[0.36]	[1.40]	[2.21]	[1.63]	[2.69]	[1.95]	[2.58]
std	0.09	0.07	0.07	0.07	0.07	0.06	0.08
skew	−0.44	−0.17	−0.11	−0.09	0.02	−0.22	0.35
kurt	0.88	0.48	0.30	0.41	0.58	0.42	0.56
SR	0.09	0.40	0.53	0.46	0.99	0.57	0.75
fd (%)	2.28	2.93	3.67	4.57	6.76		
spot (%)	1.50	0.18	0.22	1.46	0.26		
gpr (%)	0.12	0.20	0.27	0.40	0.71		
Panel B: with Transaction Costs							
mean (%)	1.26	2.37	3.00*	2.60	5.99**	3.04*	4.73**
<i>t</i> -stat	[0.57]	[1.19]	[1.92]	[1.34]	[2.46]	[1.79]	[2.14]
std	0.09	0.07	0.07	0.07	0.07	0.06	0.08
skew	−0.44	−0.18	−0.11	−0.10	0.02	−0.22	0.35
kurt	0.88	0.48	0.30	0.41	0.58	0.43	0.55
SR	0.14	0.35	0.46	0.39	0.91	0.52	0.62
fd (%)	2.29	2.91	3.66	4.50	6.71		
spot (%)	1.03	0.54	0.66	1.90	0.72		
gpr (%)	0.12	0.20	0.27	0.40	0.71		

This table displays descriptive statistics for currency portfolios organized by lagged geopolitical risk indices. The 20% with the lowest geopolitical risk indices of all available currencies is assigned to Portfolio 1 (*P1*), while the 20% with the highest geopolitical risk indices is assigned to Portfolio 5 (*P5*). *Avg.* denotes the average currency returns across the five portfolios, and *GPR* is a long–short strategy that buys *P5* and sells *P1*. This table reports the annualized average returns (*mean*), standard deviations (*std*), skewness (*skew*), kurtosis (*kurt*), the Sharpe ratio (*SR*), the average forward discount or interest rate differential relative to the US (*fd*), and the average change in the spot exchange rate (*spot*). Panel A reports excess returns excluding transaction costs, while Panel B reports excess returns including transaction costs. Newey and West (1987) *t*-statistics with Andrews (1991) optimal lag selection are presented in square brackets, and *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. This table also reports the average geopolitical risk index (*gpr*) of each portfolio. The portfolios are rebalanced on a monthly basis, and the sample period runs from June 2002 to December 2019.

month period, based on quarterly and yearly GPR indices (Table IA4 in the Internet Appendix).

3.2. Geopolitical risk-based strategy versus carry trades

We wonder if the GPR-based strategy would capture anything more than the carry trade because sorting the GPR indices produces a set of portfolios with increasing forward discounts. Furthermore, because Bekaert and Panayotov (2020) show that good carry trades have higher average returns than standard carry trades, we want to know if the GPR-based strategy encompasses information beyond good carry trades.

To investigate this, we replicate Bekaert and Panayotov's (2020) good carry trade, selecting the G5 trade, comprising only the British pound (GBP), the Swedish krona (SEK), and the US dollar (USD), as it has the highest annualized average return among their G1–G5 trades. The returns for the G5 trade are defined as the good carry factor. According to Lustig et al. (2011), the carry factor is defined as the returns for the long–short portfolio of the five carry-trade portfolios. We run time-series regressions of the five GPR portfolios' excess returns and the GPR factor on a constant and, in turn, the carry factor and the good carry factor, respectively.

Table 7 displays the regressions' least-squares estimates. The estimated alphas (α) for the carry factor (Panel A) and the good carry factor (Panel B) show a rising trend from Portfolios 1–5, indicating that these two factors do not explain the difference in excess returns across GPR portfolios. Moreover, the constant terms of the high-minus-low portfolio (i.e., the GPR factor) are positive at 0.52% (*t*-statistic of 3.07) per month in Panel A and 0.48% (*t*-statistic of 2.77) per month in Panel B, both

of which are highly statistically significant. In addition, the carry factor and good carry factor hardly capture any time-series variation in the GPR portfolios, with R^2 ranging from 0.27% to 6.53% in Panel A and from 0.10% to 6.58% in Panel B. Furthermore, as shown in Table IA5 (see Internet Appendix), the GPR-based currency investment strategy is not a good carry strategy as proxied by Bekaert and Panayotov's (2020) G1–G5 trades. Thus, these results imply that the GPR-based strategy contains information beyond carry and good carry trades.

4. Asset pricing test

This section provides evidence that the GPR factor is significantly priced in broad currency cross sections by presenting cross-sectional asset pricing tests for currency portfolios. Furthermore, even after controlling for a number of existing risk factors, this pricing power remains strong.

4.1. Methodology

We denote $RX_{i,t+1}$ as the excess returns of portfolio *i* at time *t* + 1. In the absence of arbitrage opportunities, risk-adjusted CERs have a zero price and satisfy the basic Euler equation

$$E_t [M_{t+1} RX_{i,t+1}] = 0, \quad (4)$$

with a stochastic discount factor (SDF) linear in the pricing factors f_{t+1} , given by

$$M_{t+1} = 1 - b' (f_{t+1} - \mu). \quad (5)$$

Table 6
Geopolitical Risk Portfolios: Using Different Formula of Currency Returns.

	P1	P2	P3	P4	P5	Avg.	GPR _{Formula}
Panel A: without Transaction Costs							
mean (%)	1.95	3.57*	4.13***	3.66**	6.68***	4.00**	4.74**
t-stat	[0.94]	[1.85]	[2.74]	[2.00]	[2.96]	[2.45]	[2.33]
std	0.08	0.07	0.06	0.07	0.06	0.06	0.07
skew	−0.29	−0.10	−0.07	−0.07	0.02	−0.16	0.24
kurt	0.66	0.42	0.28	0.37	0.53	0.37	0.43
SR	0.23	0.53	0.64	0.56	1.06	0.71	0.66
fd (%)	2.26	2.87	3.53	4.39	6.33		
spot (%)	−0.31	0.69	0.60	−0.73	0.35		
gpr (%)	0.12	0.20	0.27	0.40	0.71		
Panel B: with Transaction Costs							
mean (%)	2.42	3.19*	3.68**	3.16*	6.18***	3.73**	3.76*
t-stat	[1.16]	[1.66]	[2.44]	[1.73]	[2.72]	[2.28]	[1.84]
std	0.08	0.07	0.06	0.07	0.06	0.06	0.07
skew	−0.29	−0.10	−0.07	−0.07	0.02	−0.16	0.23
kurt	0.66	0.42	0.28	0.38	0.54	0.38	0.42
SR	0.29	0.48	0.57	0.49	0.98	0.66	0.52
fd (%)	2.54	2.64	3.26	4.07	6.02		
spot (%)	−0.12	0.56	0.42	−0.91	0.16		
gpr (%)	0.12	0.20	0.27	0.40	0.71		

This table displays descriptive statistics for currency portfolios organized by lagged geopolitical risk indices. Currency excess returns are calculated using the formula $RX_{i,t+1} = (S_{i,t+1} - F_i)/S_i$, where F_i and S_i are the level of the forward and spot exchange rate in units of US dollars per foreign currency, respectively. The 20% with the lowest geopolitical risk indices of all available currencies is assigned to Portfolio 1 (P1), while the 20% with the highest geopolitical risk indices is assigned to Portfolio 5 (P5). Avg. denotes the average currency returns across the five portfolios, and $GPR_{Formula}$ is a long-short strategy that buys P5 and sells P1. This table reports the annualized average returns (*mean*), standard deviations (*std*), skewness (*skew*), kurtosis (*kurt*), the Sharpe ratio (*SR*), the average forward discount or interest rate differential relative to the US (*fd*), and the average change in the spot exchange rate (*spot*). Panel A reports excess returns excluding transaction costs, while Panel B reports excess returns including transaction costs. Newey and West (1987) *t*-statistics with Andrews (1991) optimal lag selection are presented in square brackets, and *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. This table also shows each portfolio's average geopolitical risk index (*gpr*). The portfolios are rebalanced on a monthly basis, and the sample period runs from June 2002 to December 2019.

Here, b denotes the vector of factor loadings and μ denotes factor means. This specification implies a beta-pricing model

$$E[RX_i] = \lambda' \beta_i, \quad (6)$$

where the expected excess returns of portfolio i depend on factor risk prices λ and risk quantities β_i , which are the regression betas of portfolio excess returns on the risk factors for each portfolio i . λ is related to the SDF loadings b through $\lambda = \sum_f b$, where $\sum_f = E[(f_i - \mu)(f_i - \mu)']$ denotes the variance-covariance matrix of the factors.

We estimate portfolio betas and factor risk prices in Eq. (6) using the traditional Fama and MacBeth (1973) two-pass regression method, henceforth FMB. In the first stage, we perform a time-series regression of each portfolio's excess returns on constant and proposed risk factors as follows:

$$RX_{i,t} = a_i + f_t' \beta_i + \varepsilon_{i,t}, \text{ for each } i = 1, \dots, n, \forall t \in T. \quad (7)$$

Here, $RX_{i,t}$ is the excess returns of portfolio i in month t . f_t is a $k \times 1$ vector of risk factors. T is the sample size. n is the number of portfolios being studied.

After obtaining the estimated betas of each portfolio on all risk factors ($\hat{\beta}_i$), in the second stage, we use them as explanatory variables and perform cross-sectional regression as follows:

$$\overline{RX_i} = \hat{\beta}_i' \lambda + \alpha_i, \quad i = 1, \dots, n. \quad (8)$$

Here, the left-hand side is the time-series average of the excess returns of portfolio i . λ represents the $k \times 1$ vector of factor risk prices. α_i is the pricing error of portfolio i . In line with previous research, we do not include a constant in the second stage, which means we do not allow for common overpricing or underpricing in the cross section of returns. Furthermore, we report standard errors with the Shanken (1992) adjustment to account for the error-in-variable (EIV) problem of using estimated betas in the second-stage regression.

Aside from the Fama-MacBeth regressions, the model parameters in Eq. (5) and the implied factor risk prices can be estimated using Hansen's (1982) generalized method of moments (GMM). We use the pricing errors as a set of moments and the identity-weighting matrix to implement GMM. Because the goal is to see if the model can explain the cross section of expected CERs, we only use unconditional moments and do not use instruments. We attempt to price all currency portfolios equally well by estimating the first-stage GMM using an identity-weighting matrix. Following Burnside (2011), the covariance matrix $\sum_f$ is estimated along with the other model parameters by including an additional set of corresponding moment conditions. Hence, the estimating function takes the following form:

$$g(z_t, \theta) = \begin{bmatrix} [1 - b'(f_t - \mu_f)] RX_i \\ f_t - \mu_f \\ \text{vec}((f_t - \mu_f)(f_t - \mu_f)') - \text{vec}(\sum_f) \end{bmatrix}, \quad (9)$$

where θ contains the parameters $(b', \mu_f', \text{vec}(\sum_f)')$ and z_t represents the data (RX_i, f_t) . By exploiting the moment conditions

Table 7
Time-series Regressions: Controlling for Carry Trades.

	P1	P2	P3	P4	P5	GPR
Panel A: Carry						
α	−0.20 [−0.86]	0.05 [0.26]	0.14 [0.99]	0.08 [0.42]	0.32** [2.19]	0.52*** [3.07]
β	0.25*** [3.00]	0.17** [2.44]	0.14*** [2.61]	0.20** [2.33]	0.20** [2.13]	−0.05 [−0.43]
R^2	5.51	4.19	3.27	5.54	6.53	0.27
Panel B: Good Carry						
α	0.02 [0.10]	0.19 [1.18]	0.23* [1.77]	0.24 [1.53]	0.50*** [2.74]	0.48*** [2.77]
β	0.22 [1.07]	0.19 [1.11]	0.32*** [2.64]	0.18 [1.44]	0.17** [2.03]	−0.05 [−0.29]
R^2	1.69	2.08	6.58	1.94	1.90	0.10

Estimates of time-series regressions are presented in this table. The excess returns of the five geopolitical risk portfolios and the geopolitical risk (*GPR*) factor are regressed on the carry factor's excess return (Panel A) and on the good carry factor's excess return (Panel B), respectively. The carry factor is constructed as the long–short portfolio of the five carry-trade portfolios (Lustig et al., 2011). The good carry factor (i.e., G5 trade) is constructed as in Bekaert and Panayotov (2020). We report the constant term (α , in percentages) and slope coefficient (β), as well as the R^2 (in percentages) of the regressions. We also display Newey and West (1987) t -statistics (in squared brackets) corrected for autocorrelation and heteroskedasticity with Andrews (1991) optimal lag selection. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The sample spans the years June 2002 to December 2019.

$E[g(z_t, \theta)] = 0$ defined by Eq. (9), estimation uncertainty, which is due to the fact that factor means and the covariance matrix of factors have to be estimated, is incorporated in the standard errors of factor risk prices. According to Burnside (2011), the GMM-based standard errors are computed using a variant of Den Haan and Levin's (2000) vector autoregressive heteroskedasticity- and autocorrelation-consistent (VARHAC) procedure. This point estimation method is identical to the Fama–MacBeth two-pass regression method (Burnside, 2011).

4.2. Cross-sectional regressions

To investigate the pricing performance of the GPR factor, we first use the two-stage procedure of Fama and MacBeth (1973) to estimate portfolio betas and factor risk prices. The risk factors include the GPR factor (Table 5), the dollar (*DOL*) factor, which represents the expected currency market excess returns as well as the carry (*CAR*) factor. The dollar factor is calculated as the average return on a portfolio strategy that is long in all foreign currencies with equal weights and short in the domestic currency. According to recent literature, the dollar and carry factors are common factors in the FX market (Lustig et al., 2011). Excess returns to 30 portfolios from six currency strategies: 5 GPR portfolios, 5 carry-trade portfolios, 5 short-term momentum portfolios, 5 long-term momentum portfolios, 5 value portfolios, and 5 term spread portfolios are the test assets. The use of a large cross section reduces the possibility of misleading empirical results due to the strong factor structure inherent in test asset returns (Lewellen et al., 2010), which is especially problematic in small cross sections (Della Corte et al., 2016; Colacito et al., 2020).

Table 8 presents the cross-sectional asset pricing results. We report the estimates of the market prices of risk (λ), the cross-sectional R^2 , and the square root of mean-squared errors (RMSE). The standard errors and p -values of χ^2 tests are adjusted following Shanken (1992) for the EIV problem arising from using estimated betas in the second-stage regression.

In Panel A, we primarily focus on the sign and statistical significance of the risk price of the GPR factor. We find that the estimated risk price (λ_{GPR}) for the GPR factor is positive at 0.90% per month in the two-

Table 8
Cross-sectional Asset Pricing Tests: FMB.

λ_{DOL}	λ_{CAR}	λ_{GPR}	R^2	RMSE	χ^2
Panel A: with Carry-Trade and Geopolitical Risk Portfolios					
0.22 [1.09]		0.90*** [3.45]	45.49	0.17	{0.07}
0.21 [1.04]	1.11*** [4.18]		66.21	0.13	{0.21}
0.21 [1.06]	0.96*** [3.64]	0.64** [2.56]	89.22	0.08	{0.70}
Panel B: without Carry-Trade and Geopolitical Risk Portfolios					
0.21 [1.09]		0.72*** [3.03]	17.96	0.23	{0.00}
0.19 [0.97]	1.06*** [4.13]		55.05	0.17	{0.04}
0.21 [1.06]	0.98*** [3.77]	0.57** [2.40]	88.57	0.09	{0.35}

The asset pricing results using Fama–MacBeth (FMB) regressions are reported in this table. Excess returns of 30 portfolios from six currency strategies are used as test assets in Panel A: 5 geopolitical risk portfolios, 5 carry-trade portfolios, 5 short-term momentum portfolios, 5 long-term momentum portfolios, 5 value portfolios, and 5 term spread portfolios. Panel B test assets are the excess returns of 20 portfolios after excluding the 5 geopolitical risk and 5 carry-trade portfolios from the above set of test assets. The set of pricing factors includes the geopolitical risk (*GPR*) factor, the dollar (*DOL*) factor and the carry (*CAR*) factor of Lustig et al. (2011). We report the Fama and MacBeth (1973) estimates of the market price of risk (λ), the cross-sectional R^2 and the square root of mean-squared errors (RMSE), which are all in percentage points. The t -statistics in square brackets are adjusted for the EIV problem of betas following Shanken (1992), as well as the p -values of χ^2 tests in curly braces, which tests the null that all pricing errors are jointly zero. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Excess returns are calculated as a percentage per month. The portfolios are rebalanced on a monthly basis, and the sample period runs from June 2002 to December 2019.

factor model of the DOL and GPR factors, demonstrating high statistical significance (t -statistic of 3.45). This risk price implies that an asset with a one beta earns a risk premium of 10.80% per annum. Moreover, the positive risk price of the GPR factor suggests that portfolios exhibiting positive covariance with the GPR factor require a higher risk premium, while those that comove negatively with the GPR factor obtain a lower risk premium. Besides, the DOL factor remains insignificant in all cases, implying that it does not explain any of the cross-sectional variations in the currency returns. Further, when we augment the CAR factor into the two-factor specification of the DOL and GPR factors, the GPR factor remains significantly priced, with risk prices of 0.64% (t -statistic of 2.56). In addition, the estimated risk prices of the CAR factor also show statistical significance in each case, with a risk price of 1.11% (t -statistic of 4.18) in its two-factor model and 0.96% (t -statistic of 3.64) in the three-factor model. However, our bottom line is that the GPR factor is significantly priced in cross-sectional CERs and performs well with or without the CAR factor.

We remove the carry-trade and GPR portfolios from the test assets in Panel B and run the same tests as in Panel A. It is an out-of-sample test by excluding the portfolios used to construct the GPR and CAR factors. Besides, since the CAR and GPR factors are both tradable risk factors, their risk prices must match their expected returns. Hence, we include the GPR and CAR factors as test assets to ensure arbitrage-free estimates of λ_{CAR} and λ_{GPR} (Lewellen et al., 2010). The empirical results in Panel B are consistent with those in Panel A, indicating that the pricing power of the GPR factor recorded above is not driven by the factor structure embedded in the GPR and carry-trade portfolios, but instead

Table 9
Cross-sectional Asset Pricing Tests: GMM.

b_{DOL}	b_{CAR}	b_{GPR}	λ_{DOL}	λ_{CAR}	λ_{GPR}	R^2	$RMSE$	χ^2
Panel A: with Carry-Trade and Geopolitical Risk Portfolios								
19.83*** [3.94]		26.42*** [6.70]	0.22* [1.69]		0.90*** [5.55]	45.49	0.17	{0.02}
0.80 [0.17]	20.07*** [4.42]		0.21 [1.58]	1.11*** [6.35]		66.21	0.13	{0.21}
11.67** [2.24]	16.50*** [3.72]	18.73*** [4.55]	0.21 [1.65]	0.96*** [5.55]	0.64*** [3.81]	89.22	0.08	{0.43}
Panel B: without Carry-Trade and Geopolitical Risk Portfolios								
17.33*** [3.71]		21.86*** [6.37]	0.21* [1.67]		0.72*** [4.74]	17.96	0.23	{0.00}
0.58 [0.12]	19.16*** [4.32]		0.19 [1.45]	1.06*** [6.24]		55.05	0.17	{0.05}
10.65** [2.11]	16.87*** [3.86]	16.99*** [4.49]	0.21 [1.65]	0.98*** [5.82]	0.57*** [3.66]	88.57	0.09	{0.22}

This table reports the asset pricing results using the generalized method of moments (GMM) method. The test assets and pricing factors are the same as in Table 8. We report the GMM one-step estimates of factor loadings on the pricing kernel (b) and prices of factor risk (λ). Market prices of risk (λ), the cross-sectional R^2 and the square root of mean-squared errors ($RMSE$) are reported in percentage points. The t -statistics in square brackets are computed with GMM-VARHAC standard errors, as well as the p -values of χ^2 tests in curly braces, which tests the null that all pricing errors are jointly zero. *, ** and *** denote significance at the 10%, 5%, and 1% levels, respectively. Excess returns are calculated as a percentage per month. The portfolios are rebalanced on a monthly basis, and the sample period runs from June 2002 to December 2019.

extends to other currency cross sections. Furthermore, in Table IA6 (see Internet Appendix), we report the asset pricing results for a cross section of portfolios from each of the six currency strategies and find that the pricing power of the GPR factor remains strong in all cases. These findings suggest that the GPR factor is strongly priced in a broad currency cross section comprising various factor structures, and that this pricing power is unaffected by conventional carry risks.

In addition to the FMB estimates, we present the results of asset pricing tests performed on a cross section of currency portfolios using GMM. We report GMM estimates of the factor loading b and implied λ , as well as cross-sectional R^2 and the RMSE. We also report the t -statistics computed with GMM-VARHAC standard errors, as well as the p -values of the χ^2 statistic for testing the null hypothesis that the cross-sectional pricing errors are jointly equal to zero.

We find in Table 9 that the GMM estimates of both b_{GPR} and λ_{GPR} are significantly positive regardless of whether the CAR factor is included in the model. This implies that the GPR factor is more than just a priced factor; it also helps to explain currency returns when the dollar and carry factors are present. In terms of point estimates, the first-stage GMM estimates are equivalent to the second-pass FMB regression, which is consistent with Burnside (2011). In the three-factor models, we also uncover a strong cross-sectional fit in terms of the R^2 , and we cannot reject the null that all pricing errors are jointly equal to zero. Furthermore, to address the issue of small sample biases (Bekaert and De Santis, 2021), we use bootstrap methods to construct confidence regions for the parameters. The results reported in Table IA7 (see Internet Appendix) show that the bootstrapped confidence regions associated with the GPR factor do not encompass zero and contain estimates of b_{GPR} and λ_{GPR} . These results suggest that the independent information in GPR matters for pricing the currency portfolios.

4.3. Controlling for alternative risk factors

We further investigate whether including other risk factors could attenuate the explanatory power of the GPR factor in the cross section of currency portfolios. We begin by controlling for global FX volatility (VOL) as Menkhoff et al. (2012a) described, which has explanatory power for currency carry portfolios. Second, we consider the TED spread (TED), which is a proxy for funding liquidity risk, because the

literature suggests that liquidity can help us understand the cross section of carry-trade returns (Brunnermeier et al., 2008; Menkhoff et al., 2012a). Third, we look at other indicators of uncertainty that influence asset returns. Among these are Bekaert et al.'s (2022) Bekaert–Engstrom–Xu US Risk Aversion Index (BEXRA), a new measure of time-varying risk aversion that can eventually be calculated from observable financial data at high frequencies. Furthermore, we examine the global economic policy uncertainty (EPU) developed by Baker et al. (2016), as well as the global political risk (PR) proposed by Filippou et al. (2018).¹³ We begin by computing the innovations to these variables as the residuals from a simple AR(1) for the level of these indices. Noting that these proxies are not traded portfolios, we build factor-mimicking portfolios by regressing these factor innovations on five carry portfolios, as Menkhoff et al. (2012a) do.

We first examine the relationship between the GPR factor and these alternative risk factors. We show the correlation between all factors in their innovation and factor-mimicking portfolios. In Panel A of Table 10, we find that the GPR factor is only weakly correlated with innovations to TED (0.03), EPU (0.01), and PR (−0.01), but more so with VOL (0.32) and BEXRA (0.40). In Panel B of Table 10, the GPR factor correlates more with the factor-mimicking version of TED (0.25), EPU (0.17), and PR (−0.32) than with their innovations. The correlations of the GPR factor with the factor-mimicking version of VOL (0.37) and BEXRA (0.45) are close to those with their innovations. The weak

¹³ The global FX volatility is constructed as in Menkhoff et al. (2012a) using daily exchange rate quotes from WM/Reuters via Datastream. Federal Reserve Economic Data provides data for the TED spread. Data for the Bekaert–Engstrom–Xu US Risk Aversion Index are available from <https://www.nancyxu.net/risk-aversion-index>. Data for the monthly global economic policy uncertainty of Baker et al. (2016) are available from http://www.policyuncertainty.com/global_monthly.html. Baker et al. (2016) present two versions of the global economic policy uncertainty index, one based on current-price GDP and the other on PPP-adjusted GDP. The empirical results using these two measures are very similar, and in this section we present the results using the one based on current-price GDP measures. The proxy for global political risk relative to the political situation in the United States is built using monthly country-level political risk score data from the International Country Risk Guide (ICRG) database, as described by Filippou et al. (2018). All variable definitions are in Appendix Table A.1.

Table 10
Correlation Matrix.

	GPR	VOL	TED	BEXRA	EPU	PR
Panel A: Innovations						
GPR	1					
VOL	0.32	1				
TED	0.03	0.10	1			
BEXRA	0.40	0.34	0.20	1		
EPU	0.01	0.30	0.19	0.16	1	
PR	−0.01	0.02	0.08	−0.01	−0.06	1
Panel B: Factor-Mimicking Portfolios						
GPR	1					
VOL	0.37	1				
TED	0.25	0.88	1			
BEXRA	0.45	0.89	0.67	1		
EPU	0.17	0.74	0.77	0.53	1	
PR	−0.32	−0.55	−0.30	−0.54	−0.11	1

This table presents correlation coefficients of the geopolitical risk (*GPR*) factor with a series of risk factors, including the innovations (Panel A) and factor-mimicking portfolios (Panel B) of the global FX volatility (*VOL*) from Menkhoff et al. (2012a), the TED spread (*TED*) used in Brunnermeier et al. (2008), the Bekaert–Engstrom–Xu US Risk Aversion Index (*BEXRA*) from Bekaert et al. (2022), the global economic policy uncertainty (*EPU*) from Baker et al. (2016), and the global political risk (*PR*) from Filippou et al. (2018). The residuals from a simple AR(1) for the level of these indices are used to compute the innovations to these variables. Factor-mimicking portfolios are constructed by regressing factor innovations on five carry-sorted portfolios as in Menkhoff et al. (2012a). All factors occur on a monthly basis. The sample spans the years June 2002 to December 2019.

correlation of the GPR factor with these alternative risk factors indicates that these indicators contain different aspects of information.

Then, using the aforementioned factors as control variables, we run cross-sectional asset pricing tests. We control for the dollar and carry factor in all regressions. Table 11 displays the Fama–MacBeth regression estimates for 30 test portfolios (as above). Regardless of the control variable, we find that the estimated risk prices of the GPR factor are consistently positive and statistically significant. All specifications demonstrate satisfactory explanatory power with R^2 above 89%. The p -values of the χ^2 tests are above 10%, suggesting that we cannot reject the hypothesis that pricing errors are jointly zeros for each case. Overall, empirical results suggest that the GPR factor is effective at pricing cross-sectional CERs. This pricing power remains strong even after controlling for various existing risk factors and test assets.

4.4. Portfolios based on betas

We examine GPR's pricing power from a different angle. If the GPR factor is a priced risk factor, then currencies sorted by ex-ante exposure to the GPR factor should produce a cross section of portfolios with an economically meaningful spread in average returns. Specifically, for each time t , we first regress each currency's excess return on a constant and the GPR factor using a 36-month rolling window that ends in period $t - 1$, and obtain currency k 's exposure to the GPR factor, denoted as $\beta_{GPR,t}^k$. We then sort currencies into five portfolios based on $\beta_{GPR,t}^k$ at time t , where Portfolio 1 (Portfolio 5) contains 20% of the most negatively (positively) exposed currencies.

The descriptive statistics for the five beta-sorted portfolios are shown in Table 12. The average returns have a similar return distribution to the GPR portfolios shown in Table 5. Before and after considering transaction costs, the five portfolio returns tend to increase when moving from Portfolio 1 (1.17% and 1.28% per annum, respectively) to Portfolio 5 (5.11% and 5.07% per annum, respectively). Hence, the zero-cost strategy that is long Portfolio 5 and short Portfolio 1 yields an

Table 11
Cross-sectional Asset Pricing Tests: Alternative factors.

	VOL	TED	BEXRA	EPU	PR
λ_{GPR}	0.64** [2.55]	0.64** [2.57]	0.64** [2.55]	0.64** [2.55]	0.63** [2.55]
$\lambda_{CONTROL}$	−0.01*** [−2.71]	−1.81** [−2.47]	−4.05 [−1.46]	−2.11** [−2.38]	0.03 [1.07]
λ_{DOL}	0.21 [1.04]	0.21 [1.05]	0.21 [1.04]	0.21 [1.06]	0.21 [1.05]
λ_{CAR}	0.96*** [3.62]	0.95*** [3.62]	0.98*** [3.67]	0.96*** [3.62]	0.98*** [3.68]
R^2	89.74	89.32	89.61	89.24	89.56
$RMSE$	0.07	0.07	0.07	0.08	0.07
χ^2	{0.71}	{0.68}	{0.69}	{0.65}	{0.66}

This table reports the asset pricing results after including other control factors in addition to the geopolitical risk (*GPR*) factor, the dollar (*DOL*) factor, and the carry (*CAR*) factor. The control factors include the factor-mimicking portfolios of the global FX volatility (*VOL*) from Menkhoff et al. (2012a), the TED spread (*TED*) used in Brunnermeier et al. (2008), the Bekaert–Engstrom–Xu US Risk Aversion Index (*BEXRA*) from Bekaert et al. (2022), the global economic policy uncertainty (*EPU*) from Baker et al. (2016), and the global political risk (*PR*) from Filippou et al. (2018). The residuals from a simple AR(1) for the level of these indices are used to compute the innovations to these variables. Factor-mimicking portfolios are constructed by regressing factor innovations on five carry-sorted portfolios as in Menkhoff et al. (2012a). The excess returns of 30 portfolios from six currency strategies are used as test assets: five geopolitical risk portfolios, five carry-trade portfolios, five short-term momentum portfolios, five long-term momentum portfolios, five value portfolios, and five term spread portfolios. We report the Fama and MacBeth (1973) estimates of the market price of risk (λ), the cross-sectional R^2 and the square root of mean-squared errors ($RMSE$), which are all in percentage points. The t -statistics in square brackets are adjusted for the EIV problem of betas following Shanken (1992), as well as the p -values of χ^2 tests in curly braces which tests the null that all pricing errors are jointly zero. *, ** and *** denote significance at the 10%, 5%, and 1% levels, respectively. Excess returns are expressed in percentage per month. The portfolios are rebalanced monthly, and the sample runs from June 2002 to December 2019.

annual excess return of 3.94% before transaction costs and 3.79% after costs, which are both statistically significant. We also observe that the postformation betas, obtained by regressing the realized excess returns of the beta-sorted portfolios on a constant and the GPR factor, increase from Portfolio 1 to Portfolio 5. Table IA8 also reports the results using a 24-month rolling window (see Internet Appendix). The results are improved with the long–short portfolio exhibiting an excess return of 4.55% (t -statistic of 2.01) per annum before considering the transaction costs and 4.40% (t -statistic of 1.94) after costs. These findings confirm the strong relationship between the GPR and CERs.

5. Commonality in GPR

5.1. Estimation of common GPR measures

Based on the country-specific GPR indices, our findings so far suggest a strong relationship between GPR and cross-sectional CERs. Geopolitical events or threats rarely involve a single country. Many geopolitical events, such as the Kuwait invasion and subsequent Gulf War, are global and can spill over to multiple countries. As a result, in this section, we will continue to look into the commonality in GPR.

To test for commonality across the individual country GPR indices, we first construct a measure of the common GPR index. Specifically, we estimate the global commonality in the GPR indices in month t (i.e., $gpr_{global,t}$) simply as the cross-sectional average of the individual GPR indices ($gpr_{j,t}$) at the country level (Chordia et al., 2000; Pástor and Stambaugh, 2003; Mancini et al., 2013),

Table 12
Descriptive Statistics: Portfolios Sorted on Betas.

	P1	P2	P3	P4	P5	Avg.	GPR _{beta}
Panel A: without Transaction Costs							
mean (%)	1.17	0.72	2.31	2.10*	5.11***	2.28	3.94*
<i>t</i> -stat	[0.42]	[0.29]	[1.17]	[1.69]	[3.19]	[1.27]	[1.75]
std	0.11	0.09	0.08	0.05	0.04	0.07	0.10
skew	−0.23	−0.25	−0.17	−0.08	0.13	−0.22	0.16
kurt	0.42	0.50	0.59	0.29	0.41	0.47	0.34
SR	0.11	0.08	0.29	0.41	1.15	0.35	0.40
prefd (%)	3.87	1.85	2.20	1.88	6.31	3.22	2.44
postfd (%)	3.83	2.00	1.89	1.86	6.56	3.23	2.73
pre-β	−0.89	−0.56	−0.39	−0.20	0.10		
std	(0.38)	(0.31)	(0.28)	(0.20)	(0.19)		
post-β	−0.55	−0.02	0.05	0.08	0.45		
s.e	(0.05)	(0.05)	(0.05)	(0.03)	(0.05)		
Panel B: with Transaction Costs							
mean (%)	1.28	0.64	2.28	2.08*	5.07***	2.27	3.79*
<i>t</i> -stat	[0.46]	[0.25]	[1.16]	[1.67]	[3.17]	[1.26]	[1.68]
std	0.11	0.09	0.08	0.05	0.04	0.07	0.10
skew	−0.23	−0.25	−0.18	−0.08	0.14	−0.23	0.16
kurt	0.42	0.51	0.59	0.29	0.41	0.48	0.34
SR	0.12	0.07	0.29	0.41	1.14	0.34	0.38
prefd (%)	3.74	1.80	2.21	1.87	6.29	3.18	2.56
postfd (%)	3.83	2.01	1.89	1.86	6.54	3.23	2.71
pre-β	−0.89	−0.56	−0.39	−0.20	0.10		
std	(0.38)	(0.31)	(0.28)	(0.20)	(0.19)		
post-β	−0.56	−0.02	0.05	0.08	0.44		
s.e	(0.05)	(0.05)	(0.05)	(0.03)	(0.05)		

The table presents descriptive statistics of β -sorted currency portfolios. Each β at time t is obtained by regressing individual currency excess returns on the geopolitical risk factor using a 36-month rolling window that ends in period $t - 1$. Portfolio 1 (P1) contains the 20% of currencies with the lowest betas, and Portfolio 5 (P5) contains the 20% of currencies with the highest betas. Avg. denotes the average currency returns across the five portfolios and GPR_{β} is a long-short strategy that buys P5 and sells P1. Panel A reports excess returns excluding transaction costs, while Panel B reports excess returns including transaction costs. This table reports annualized average returns (*mean*), standard deviations (*std*), skewness (*skew*), kurtosis (*kurt*), and Sharpe ratios (*SR*). There are also the average preformation (*prefd*) and postformation (*postfd*) forward discounts for each portfolio (in % per annum), preformation betas (*pre-β*) with standard deviations in parentheses, and postformation (*post-β*) betas with standard errors in parentheses for each portfolio. The *t*-statistics of Newey and West (1987) with Andrews (1991) optimal lag selection are presented in square brackets, and *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The portfolios are rebalanced on a monthly basis, and the sample period runs from June 2002 to December 2019.

$$gpr_{global,t} = \frac{1}{n_t} \sum_{j=1}^{n_t} gpr_{j,t}, \quad (10)$$

where n_t is the total number of available countries in time t . We also compute the regional commonality in GPR indices, denoted as $gpr_{regional,t}$, as the equally weighted cross-sectional average of the individual GPR indices for all countries in that region (Ismailescu and Phillips, 2015).¹⁴

We then investigate how individual country GPR indices relate to global and regional GPR indices. We run time-series regressions of the country-specific GPR index ($gpr_{j,t}$) on the global GPR index ($gpr_{global,t}$) and regional GPR index ($gpr_{regional,t}$) for country j in the manner of Chordia et al. (2000) and Mancini et al. (2013), as follows:

$$gpr_{j,t} = \alpha_j + \beta_{1,j} gpr_{global,t} + \beta_{2,j} gpr_{regional,t} + gpr_{I,j,t}, \quad (11)$$

where $gpr_{I,j,t}$ represents an idiosyncratic GPR shock.¹⁵ The slope coefficients capture the sensitivity, that is, $\beta_{1,j}$ and $\beta_{2,j}$.

For each country in our sample, we run the model in Eq. (11) and report the cross-sectional averages of the slope coefficients across countries in Table 13. The average sensitivities to the global GPR index (i.e., β_1) and the regional GPR index (i.e., β_2) across countries are positive and highly significant in all cases. In the one-factor models, more than 92% of these individual slope coefficients (i.e., β_1 and β_2) are positive and more than 59% are statistically significant at the 5% significance level. Furthermore, in the two-factor models, the average β_1 is 0.25 with a *t*-statistic of 2.22, while the average β_2 is 0.68 with a *t*-statistic of 4.80. This is consistent with our intuition; that is, in contrast to global geopolitical turmoil, a country's GPR is more influenced by geopolitical

¹⁴ We use five broad regions classified by the United Nations (UN): *African States*, *Asia-Pacific States*, *Eastern European States*, *Latin American and Caribbean States*, and *Western European and other States*. The United Nations Regional Groups are geopolitical regional groups of United Nations member states. The classification is available from <https://www.un.org/dgacm/en/content/regional-groups>.

¹⁵ To avoid potentially upward-biased sensitivities, we exclude country j in the computation of $gpr_{global,t}$ and $gpr_{regional,t}$ (Mancini et al., 2013; Verdelhan, 2018). Furthermore, to avoid being swayed by extreme values, we exclude the countries with the highest and lowest GPR indices when calculating commonality (Banti et al., 2012; Mancini et al., 2013).

Table 13
Time-series Regressions: Commonality in Geopolitical Risk.

	α	β_1	β_2	R^2
	0.07*** [5.55]	0.86*** [6.53]		7.17
% positive	76.92	94.23		
% significant	38.46	71.15		
	0.10*** [7.35]		0.76*** [7.27]	8.15
% positive	92.31		92.31	
% significant	63.46		59.62	
	0.08*** [6.22]	0.25** [2.22]	0.68*** [4.80]	10.45
% positive	80.77	65.38	73.08	
% significant	34.62	28.85	26.92	

This table presents the time-series regression results. We report the cross-sectional averages of the intercepts (α) and slope coefficients (β_1 and β_2) across countries, along with the percentage of positive estimates (% positive) and the percentage of estimates that are statistically significant at the 5% level (% significant) for individual countries. The cross-sectional average estimates of R^2 and intercepts are in percentage points. We also show Newey and West (1987) t -statistics (in square brackets) corrected for autocorrelation and heteroskedasticity using Andrews (1991) optimal lag selection, with *, **, and *** denoting significance at 10%, 5%, and 1% levels, respectively. The sample spans the years October 1983 to December 2019.

events specific to that country's region. Furthermore, the alphas in the various model specifications are all statistically different from zero, implying that country-specific GPR indices contain more information than global and regional GPR indices. These results imply that GPR is influenced by shocks that affect individual countries as well as shocks that affect the world or region as a whole.

5.2. Sources of GPR predictability for currency returns

As seen in the preceding results, exposures to global GPR and regional GPR, as well as idiosyncratic GPR, may be sources of GPR return predictability for currency returns. But which of these three sources of risk is truly important?

To answer this question, we divide the monthly country-specific GPR indices into three components: global risk, regional risk, and country-specific idiosyncratic risk. We assume that the global (regional) component would be determined by the realization of the global (regional) GPR index value and the country-specific GPR index's exposure to the global (regional) GPR index. The idiosyncratic risk represents the residual part of total risk that cannot be attributed to the global or regional share.

Specifically, for each month t , we regress the country-specific GPR index on the global GPR index and regional GPR index for each country, as in Eq. (11), using a 36-month rolling window. The realizations of the global and regional GPR index values in month t are then used in conjunction with the individual country exposures estimated in the rolling regressions to compute the regression-based value of the global risk component (i.e., $gpr_{j,t}^{global\ component}$) and the regional risk component (i.e., $gpr_{j,t}^{regional\ component}$). Further, the idiosyncratic risk component for each country j in month t (i.e., $gpr_{j,t}^{idiosyncratic\ component}$) is calculated as the difference between the actual country-specific GPR index value and the regression-based value.

We then divide currencies into five portfolios based on these three risk components, and the results are shown in Table 14. Portfolios sorted by country-specific idiosyncratic risk and regional risk component find an increasing pattern from Portfolio 1 to Portfolio 5, as do their corresponding long-short portfolios (i.e., $GPR_{component}$) exhibit positive excess returns with high statistical significance (4.13% with a t -statistic of 2.23 and 4.42% with a t -statistic of 2.16, respectively).

Meanwhile, the portfolios built on the global risk component show no cross-sectional pattern in returns, and the average returns on the long-short portfolio are not statistically different from zero. Hence, we can conclude that the GPR's observed return predictability for currency returns is derived from the country-specific idiosyncratic risk component and the regional risk component.

This finding is not surprising given that GPR primarily focuses on interstate affairs, encompassing a country's geographic scope and neighboring countries. Furthermore, the country-specific GPR indices are largely derived from local news sources in the Access World News database. For example, the total number of GPR-related news items retrieved for Australia from 1985 to 2019 in the Access World News database was 2,120. The highest number was 1,250 from Australia, followed by 197 from the United States. In terms of continents, Australia/Oceania had the most entries with 1,345 total, whereas North America had the second most, with only 296. As a result, the country-specific idiosyncratic and regional risk components of the country-specific GPR index are the primary source of differences in GPR indices across countries.

Furthermore, because we are focusing on cross-sectional currency returns, their variation is primarily influenced by differences in cross-sectional characteristics between countries. Global risk factors always affect all countries, while regional and country-specific risk factors exhibit greater cross-sectional variations in the effects on individual countries (Zaremba et al., 2022). As shown in Table 13, when both global and regional GPR indices are considered in the regression, the average sensitivity to the global GPR index (i.e., β_1) across countries is 0.25, which is much lower than the average sensitivity to the regional GPR index (i.e., β_2), which is 0.68. This suggests that each country is much more vulnerable to, and thus affected by, regional GPR than global GPR. Moreover, the standard deviation of β_2 across countries (1.13) is higher than that of β_1 across countries (0.83), which suggests that the cross-sectional variation in different countries' exposure to regional GPR is greater than that to global GPR. This variation implies differences in GPR profiles among cross-sectional countries. These findings are consistent with the fact that when we perform portfolio analysis based on the global and regional risk components of each country's GPR index, the regional risk component has pricing power, not the global risk component.

Furthermore, by filtering out GPR commonalities, we test the pricing power of the idiosyncratic GPR. We perform 36-month rolling window regressions of the country-specific GPR index on the global and regional GPR indices for each country, as described earlier. The residuals from these regressions are used to calculate the idiosyncratic risk. By excluding the global GPR commonality only (i.e., $gpr_{j,t}^{idioG}$), the regional GPR commonality only (i.e., $gpr_{j,t}^{idioR}$), and both the global and regional GPR commonalities (i.e., $gpr_{j,t}^{idioGR}$), we obtain three measures of idiosyncratic GPR. The currencies are then divided into five portfolios based on these three idiosyncratic GPR measures, and the corresponding idiosyncratic GPR factors are calculated by subtracting Portfolio 1 from Portfolio 5, labeled as GPR_{idioG} , GPR_{idioR} , and GPR_{idioGR} .

We then run the cross-sectional asset pricing tests using the DOL factor and the CAR factor from Lustig et al. (2011) and these three idiosyncratic GPR factors. The results in Table 15 show that the estimated risk prices of the three idiosyncratic GPR factors remain highly significantly positive in all cases, even after controlling for the CAR factor. The risk price estimates range from 0.35% to 0.92%, which are similar to the estimates when using the original GPR measure in the baseline results. As a result, even after removing the GPR commonalities, the explanatory power of GPR in the cross section of currency returns remains robust.

6. Robustness tests

6.1. Alternative GPR measures

This section considers three different GPR measures. To begin, Access World News offers 11 different source types, including *Audio*, *Blog*,

Table 14
Descriptive Statistics: Portfolios Sorted on Risk Components.

	P1	P2	P3	P4	P5	Avg.	GPR _{component}
Panel A: Idiosyncratic Risk Component							
mean	3.67**	2.66	3.25*	2.21	7.81***	3.92***	4.13**
<i>t</i> -stat	[2.39]	[1.56]	[1.65]	[1.14]	[3.54]	[2.64]	[2.23]
std	5.53	6.94	8.47	7.52	7.12	5.42	7.28
SR	0.66	0.38	0.38	0.29	1.10	0.72	0.57
Panel B: Global Risk Component							
mean	3.20**	3.49**	−1.18	3.20*	3.82*	2.51*	0.61
<i>t</i> -stat	[2.42]	[2.19]	[−0.77]	[1.66]	[1.82]	[1.88]	[0.36]
std	5.71	6.76	7.46	7.55	7.26	5.49	6.82
SR	0.56	0.52	−0.16	0.42	0.53	0.46	0.09
Panel C: Regional Risk Component							
mean	0.43	0.73	1.54	4.78**	4.85**	2.47*	4.42**
<i>t</i> -stat	[0.33]	[0.45]	[0.87]	[2.18]	[2.10]	[1.80]	[2.16]
std	6.44	7.71	7.23	7.32	7.26	5.47	7.45
SR	0.07	0.09	0.21	0.65	0.67	0.45	0.59

This table displays descriptive statistics for currency portfolios organized by the three geopolitical risk components, namely the idiosyncratic risk component (Panel A), the global risk component (Panel B), and the regional risk component (Panel C). The 20% with the lowest geopolitical risk component values of all available currencies is assigned to Portfolio 1 (*P1*), while the 20% with the highest geopolitical risk component values is assigned to Portfolio 5 (*P5*). Avg. denotes the average currency returns across the five portfolios, and $GPR_{component}$ is a long–short strategy that buys *P5* and sells *P1*. This table reports the annualized average returns (*mean*, in percentages), standard deviations (*std*) and the Sharpe ratios (*SR*). Newey and West (1987) *t*-statistics with Andrews (1991) optimal lag selection are presented in square brackets, and *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The portfolios are rebalanced on a monthly basis. The sample spans the years October 1983 to December 2019.

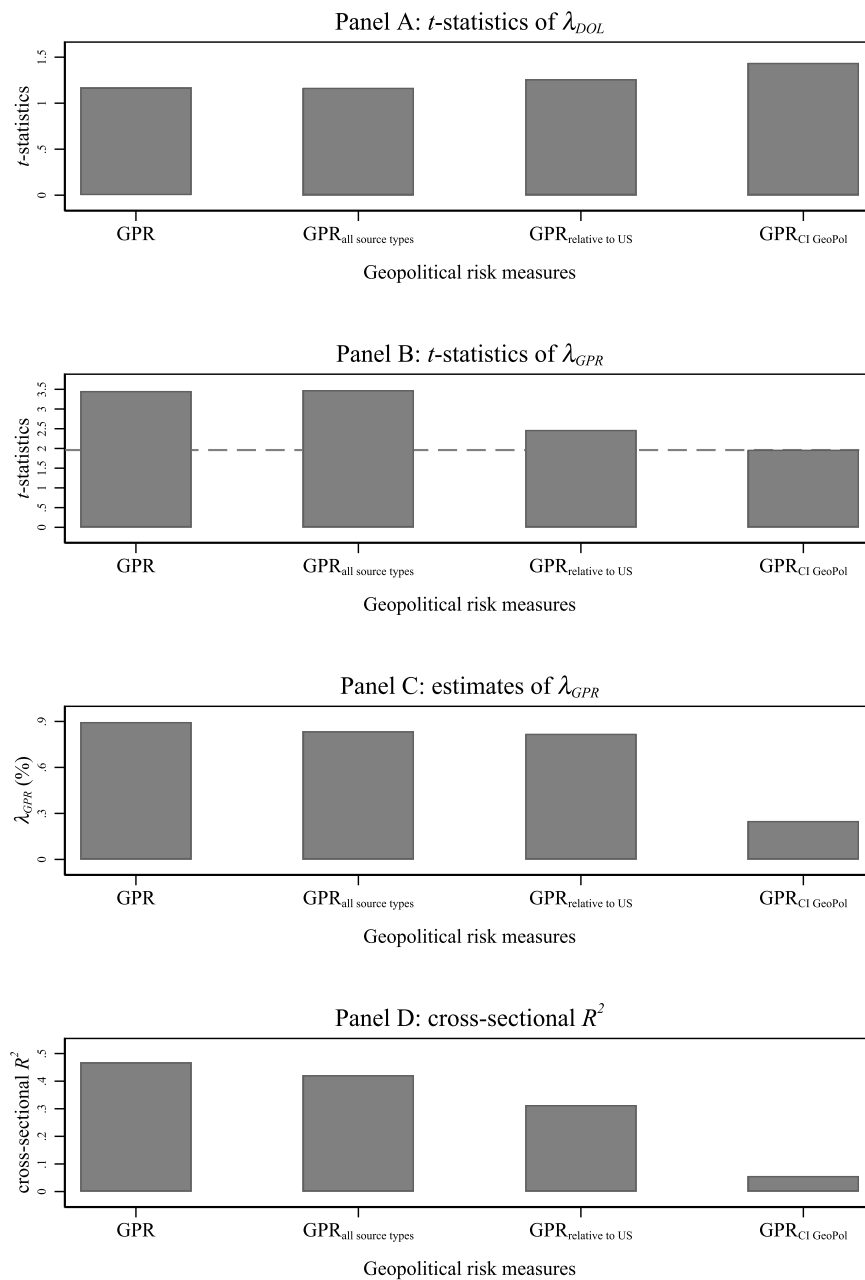
Table 15
Asset Pricing Tests: Idiosyncratic Geopolitical Risk Measures.

λ_{DOL}	λ_{CAR}	$\lambda_{GPR_{idioG}}$	$\lambda_{GPR_{idioR}}$	$\lambda_{GPR_{idioGR}}$	R^2	$RMSE$	χ^2
0.11 [0.64]		0.60*** [3.06]			12.21	0.21	{0.00}
0.11 [0.66]	1.10*** [4.72]	0.35* [1.81]			78.32	0.10	{0.11}
0.22 [1.19]			0.92*** [3.87]		22.88	0.21	{0.00}
0.22 [1.18]	1.06*** [4.10]		0.58*** [2.61]		63.85	0.14	{0.01}
0.22 [1.20]				0.73*** [3.35]	11.67	0.22	{0.00}
0.22 [1.18]	1.08*** [4.14]			0.52** [2.41]	65.87	0.14	{0.02}

This table presents the cross-sectional asset pricing results using three different idiosyncratic geopolitical risk factors, that is, GPR_{idioG} , GPR_{idioR} , and GPR_{idioGR} . The set of pricing factors also includes the dollar (*DOL*) factor and the carry (*CAR*) factor from Lustig et al. (2011). The 5 geopolitical risk portfolios under each idiosyncratic geopolitical risk measure, as well as the 5 carry-trade portfolios, 5 short-term momentum portfolios, 5 long-term momentum portfolios, 5 value portfolios, and 5 term spread portfolios, are all included in the test assets. We present Fama and MacBeth (1973) estimates of the risk market price (λ), the cross-sectional R^2 and the square root of mean-squared errors ($RMSE$) in percentage points. The *t*-statistics in square brackets are adjusted for the EIV problem of betas following Shanken (1992), as well as the *p*-values of χ^2 tests in curly braces, which tests the null that all pricing errors are jointly zero. The symbols *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Excess returns are calculated as a percentage per month. The portfolios are rebalanced on a monthly basis, and the sample period spans October 1983 to December 2019.

College/University Newspaper, Journal, Magazine, Newsletter, Newspaper, Newswire, Transcript, Video, Web-Only Source. Instead of focusing only on the newspaper category as in the above results, we collect news items

from all source types in the Access World News database and construct GPR indices, which are labeled as $gpr_{j,t}^{all\ source\ types}$ for each country *j* in month *t*. Second, following Filippou et al. (2018), we build GPR



This graph depicts the cross-sectional asset pricing results for the four different geopolitical risk (GPR) measures. The set of risk factors include the dollar (*DOL*) factor, the *GPR* factor, which is the main measure used in the above results (see Table 5), as well as the *GPR_{all source types}* factor constructed with data from all source types in Access World News, the *GPR_{relative to US}* factor constructed being relative to the US geopolitical conditions, and the *GPR_{CI GeoPol}* factor constructed by utilizing the country-specific GPR indices of Caldara and Iacoviello (2022). Under each measure, test assets include five GPR portfolios, as well as the five carry-trade portfolios, five short-term momentum portfolios, five long-term momentum portfolios, five value portfolios, and five term spread portfolios. We report in percentage points Fama and MacBeth (1973) estimated risk prices of GPR factors (Panel C), the t -statistics of risk price estimates of the *DOL* factor (Panel A) and GPR factors (Panel B), and the cross-sectional R^2 (Panel D). The t -statistics are adjusted for the EIV problem of betas as described by Shanken (1992). The horizontal dashed line represents the significance bound of 1.96. Excess returns are calculated as a percentage per month. The portfolios are rebalanced on a monthly basis, and the sample period runs from June 2002 to December 2019.

Fig. 2. Alternative Geopolitical Risk Measures.

indices relative to US geopolitical conditions to reflect differences in geopolitical uncertainty between the US and the rest of the world. The relative measure of GPR for each country j in month t is calculated as follows:

$$gpr_{j,t}^{relative\ to\ US} = \frac{gpr_{j,t} - gpr_{US,t}}{\sigma_{j,t}}, \quad (12)$$

where $\sigma_{j,t} = (1/n_t) \sum_1^{n_t} |gpr_{j,t} - gpr_{US,t}|$, n_t is the total number of available countries in time t . Third, we utilize the country-specific GPR indices developed by Caldara and Iacoviello (2022), labeled as *CI GeoPol*.¹⁶ Further, we sort currencies into five portfolios by these three alternative measures (i.e., $gpr_{j,t}^{all\ source\ types}$, $gpr_{j,t}^{relative\ to\ US}$, $gpr_{j,t}^{CI\ GeoPol}$), respectively, and then get the corresponding GPR factors by taking the difference of Portfolio 5 and Portfolio 1, labeled as $GPR_{all\ source\ types}$, $GPR_{relative\ to\ US}$, $GPR_{CI\ GeoPol}$.

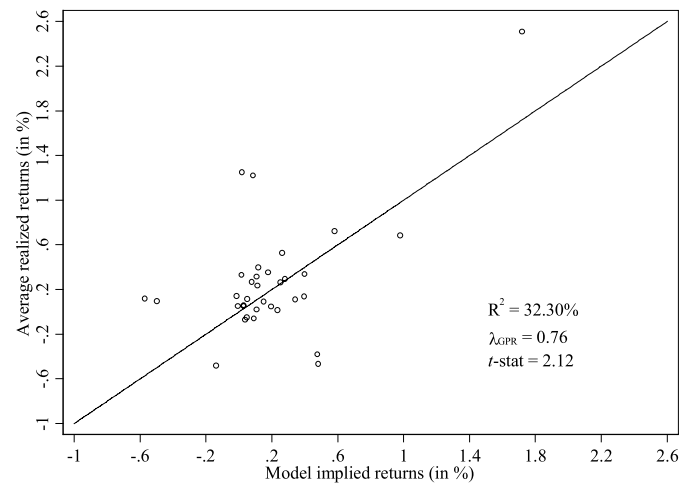
The DOL factor, the original GPR factor (see Table 5), and these three alternative measures of GPR factors are used in the cross-sectional asset pricing tests. The GPR portfolios for each measure, as well as the 5 carry-trade portfolios, 5 short-term momentum portfolios, 5 long-term momentum portfolios, 5 value portfolios, and 5 term spread portfolios, are all test assets. We report in Fig. 2 the estimates and t -statistics of factor risk prices as well as the cross-sectional R^2 . We find that the estimated risk prices of these four GPR factors are all highly significantly positive, with t -statistics ranging from 1.97 to 3.48. Moreover, the risk price estimates of the GPR and $GPR_{all\ source\ types}$ factors are very close with similar statistical significance, indicating that the original GPR measure captures sufficient information compared with the one used all source types of news. Besides, the models with $GPR_{relative\ to\ US}$ and $GPR_{CI\ GeoPol}$ demonstrate weaker cross-sectional explanatory ability in terms of cross-sectional R^2 in contrast with the other two models. Overall, the model with the original GPR factor outperforms other models using alternative measures of GPR in terms of statistical significance, estimates of factor risk prices, and cross-sectional explanatory power.

6.2. Currency-level asset pricing tests

The grouping of individual currencies into portfolios for asset pricing tests is one source of concern about the validity of our results. Classifying asset returns into portfolios has the potential to introduce data-snooping biases (Lo and MacKinlay, 1990) and destroy information due to the shrinkage of beta dispersions (Ang et al., 2020).

Hence, to address this concern, we use the currency-level excess returns as test assets and take the GPR and DOL factors as risk factors to conduct cross-sectional asset pricing tests. We use the Fama and MacBeth (1973) two-pass procedure, as recommended by Lustig et al. (2011), due to the unbalanced panel of data of individual currencies, and evaluate performance using unconditional and conditional CERs, respectively. The unconditional CERs (i.e., RX_{t+1}) are calculated as in Eq. (2). As in Della Corte et al. (2016), we calculate the conditional CERs as $RX_{t+1}^c = \gamma \times RX_{t+1}$, where $\gamma = 1$ ($\gamma = -1$) when the interest rate spread between foreign currency and the US dollar is positive (negative). Furthermore, in our sample period, we only include currencies with at least 24 observations, yielding 34 currencies for regression.

In Fig. 3, we show the individual currency-level pricing errors using conditional excess returns, and in Fig. IA2 we show them using unconditional excess returns (see Internet Appendix). Fig. 3 indicates that the estimated risk price of the GPR factor is positive at 0.76% per month and statistically significant with a t -statistic of 2.12, which is similar to the estimates in previous results. Moreover, the cross-sectional R^2



The pricing errors from the Fama and MacBeth (1973) two-pass regression with the conditional excess returns of 34 individual currencies as test assets are shown in this figure. The risk factors are the dollar (*DOL*) factor and the geopolitical risk (*GPR*) factor. Errors are deviations from the 45° line. We also report the estimated risk price of the *GPR* factor (λ_{GPR}) in percentage points, with t -statistics based on Newey and West (1987) standard errors with Andrews (1991) optimal lag selection, as well as the cross-sectional R^2 . Excess returns are calculated as a percentage per month. The sample spans the years June 2002 to December 2019.

Fig. 3. Currency-level Pricing Errors: Conditional Excess Returns.

is 32.30%, which is satisfactory since individual CERs are noisier than portfolio returns. As shown in Fig. 3, most currencies fit well or are very close to the 45° line between the realized and model-implied conditional excess returns. Furthermore, the results in Fig. IA2 show the same pattern as in Fig. 3 and are even slightly improved. Even when individual currencies are considered, the GPR factor is significantly priced in cross-sectional CERs.

6.3. Alternative base currencies

We use four alternative non-US dollar base currencies in the portfolio construction and asset pricing tests, following Della Corte et al. (2016), namely the British pound (GBP), the Japanese yen (JPY), the Swiss franc (CHF), and the euro (Deutsche mark before 1999, DEM-EUR).

We build GPR portfolios by categorizing currencies based on their country-specific GPR indices and computing excess returns for each alternative base currency. In Table 16, we present the descriptive statistics for the portfolios and find that the average excess returns increase from Portfolio 1 to Portfolio 5 in all cases. Further, the long-short portfolios (i.e., $GPR_{non-US\ base}$) exhibit highly significant positive excess returns, ranging from 5.40% to 6.30% with t -statistics ranging from 2.49 to 2.60. The returns to these long-short portfolios are denoted as factors for each alternative base currency, labeled as GPR_{GBP} , GPR_{JPY} , GPR_{CHF} , GPR_{EUR} . These results suggest that GPR portfolios with non-US dollar base currencies have return characteristics similar to our baseline results. Furthermore, we run asset pricing tests using the four different base currencies. The results in Table IA9 (see Internet Appendix) show that, even after controlling for the CAR factor, the four GPR factors are all positively priced with high statistical significance. These results suggest that the GPR factor's pricing power remains robust regardless of the base currency used.

6.4. Other robustness tests

In addition to the major robustness checks, we run a series of other robustness tests on the pricing power of GPR in the CER cross section.

¹⁶ We collected the indices from <https://www.matteoiacoviello.com/gpr-country.htm> on July 22, 2022. Ukraine and Venezuela are excluded from our empirical analysis using Caldara and Iacoviello's (2022) data due to a lack of exchange rate data.

Table 16
Geopolitical Risk Portfolios: Alternative Base Currencies.

	P1	P2	P3	P4	P5	Avg.	GPR _{non-US base}
Panel A: GBP as base currency							
mean	1.13	3.09*	3.31	2.54	6.64***	3.34*	5.51**
<i>t</i> -stat	[0.63]	[1.65]	[1.62]	[1.31]	[2.92]	[1.97]	[2.49]
std	8.58	7.63	8.34	8.36	8.78	7.24	7.66
SR	0.13	0.40	0.40	0.30	0.76	0.46	0.72
Panel B: JPY as base currency							
mean	1.09	2.77	4.12	3.85	7.39**	3.84	6.30***
<i>t</i> -stat	[0.33]	[1.04]	[1.57]	[1.44]	[2.46]	[1.44]	[2.60]
std	12.56	10.46	10.79	10.65	10.11	10.16	7.85
SR	0.09	0.26	0.38	0.36	0.73	0.38	0.80
Panel C: CHF as base currency							
mean	−0.47	1.11	1.72	1.45	4.92**	1.75	5.40**
<i>t</i> -stat	[−0.25]	[0.60]	[1.09]	[0.70]	[2.20]	[1.05]	[2.57]
std	8.63	9.09	8.52	9.03	9.66	8.04	7.30
SR	−0.06	0.12	0.20	0.16	0.51	0.22	0.74
Panel D: DEM-EUR as base currency							
mean	0.16	2.47	2.22	2.26	5.61***	2.54*	5.45***
<i>t</i> -stat	[0.09]	[1.36]	[1.25]	[1.31]	[2.83]	[1.70]	[2.59]
std	7.34	7.52	7.84	7.70	8.23	6.55	7.34
SR	0.02	0.33	0.28	0.29	0.68	0.39	0.74

This table provides descriptive statistics for geopolitical risk portfolios using non-US base currencies such as the British pound (GBP) (Panel A), Japanese yen (JPY) (Panel B), Swiss franc (CHF) (Panel C), and euro (Deutsche mark before 1999, DEM-EUR) (Panel D). The 20% with the lowest geopolitical risk index values of all available currencies is assigned to Portfolio 1 (*P1*), while the 20% with the highest geopolitical risk index values is assigned to Portfolio 5 (*P5*). Avg. denotes the average currency returns across the five portfolios, and $GPR_{non-US\ base}$ is a long–short strategy that buys *P5* and sells *P1*. This table reports the annualized average returns (*mean*, in percentages), standard deviations (*std*) and the Sharpe ratios (*SR*). Newey and West (1987) *t*-statistics with Andrews (1991) optimal lag selection are presented in square brackets, and *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The portfolios are rebalanced on a monthly basis. The sample spans the years June 2002 to December 2019.

For the sake of brevity, we present the details and results in Internet Appendix B. Our primary results remain robust after i) taking into account currency tradability by excluding countries that have both cross-border capital restrictions and fixed or quasi-fixed exchange rate regimes each month; ii) using CERs that are net of transaction costs; iii) including a risk factor related to the comovement of country-specific GPR indices with the global GPR index; iv) considering key episodes of distress in the FX market during the sample period; v) including the set of test assets currency portfolios sorted based on the output gap (Colacito et al., 2020); and vi) using the factor-mimicking portfolio for the global GPR index as an alternative GPR factor.

7. Conclusion

This study investigates the relationship between GPR and CERs. We capture GPR by focusing on news stories about geopolitical threats, nuclear threats, war threats, terrorist threats, war acts, and terrorist acts. Our research question is whether GPR can be used as a new source for the risk-based explanation of cross-sectional currency returns.

Using monthly GPR indices for 42 countries and regions from June 2002 to December 2019, we show that categorizing currencies into portfolios based on the GPR indices results in a large spread in returns. Furthermore, the GPR factor, as proxied by the long–short strategy of GPR portfolios, is positively and significantly priced in the cross section of CERs, explaining the variation in currency returns. This pricing ability extends not only to the cross section of portfolios sorted by GPR indices, carry, momentum, value, and term spreads but also to individual currencies. Furthermore, when conventional risk factors such

as carry, volatility, liquidity, political risk, and economic policy uncertainty are included in the pricing model, the GPR factor retains its explanatory power. Further investigation reveals that the observed return predictability of the GPR for currency returns stems from the GPR index's country-specific idiosyncratic risk component and regional risk component. Furthermore, after removing the GPR commonalities, the explanatory power of GPR remains robust. Finally, our primary findings remain robust after conducting a series of robustness tests.

Overall, these results suggest that GPR is a key determinant of the cross-sectional CERs, providing a new perspective on the foreign exchange market's risk-return profile.

CRedit authorship contribution statement

Xi Liu: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Xueyong Zhang:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Data availability

The authors do not have permission to share data.

Appendix. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jbankfin.2024.107097>.

Table A.1
Variable Definitions.

Variable	Definition	Source
RX_{t+1}	The excess return on buying a foreign currency in the forward market at time t and then selling it in the spot market at time $t + 1$.	Authors' calculation based on spot and forward exchange rate quotes from WM/Reuters accessed via Datastream.
$gpr_{j,t}$	The geopolitical risk index. It is computed as the proportion of news discussing geopolitical risks in the total news each month t for each country j .	Authors' calculation based on the raw news volume data crawled from Access World News.
GPR	The geopolitical risk factor. It is built as the long-short portfolio from the five geopolitical risk portfolios, which is the difference between the portfolio with the highest 20% of geopolitical risk indices and the portfolio with the lowest 20% of geopolitical risk indices.	Authors' calculation based on country-specific geopolitical risk indices.
CAR	The carry factor proposed by Lustig et al. (2011). It is built as the long-short portfolio of the five carry-trade portfolios, which is the difference between the portfolio with the highest 20% forward discount (or highest interest rate differential relative to the US) and the portfolio with the lowest 20% forward discount (or lowest interest rate differential relative to the US).	Authors' calculation based on spot and forward exchange rate quotes from WM/Reuters accessed via Datastream.
DOL	The dollar factor proposed by Lustig et al. (2011). It is calculated as the average return on a portfolio strategy in which all foreign currencies are long with equal weights and the domestic currency is short.	Authors' calculation based on spot and forward exchange rate quotes from WM/Reuters accessed via Datastream.
VOL	The global FX volatility factor. It is constructed as in Menkhoff et al. (2012a).	Authors' calculation based on daily exchange rate quotes from WM/Reuters via Datastream.
TED	The TED spread. It is calculated as the difference between the three-month USD Libor and the three-month US T-bill rate.	Federal Reserve Economic Data.
$BEXRA$	The Bekaert-Engstrom-Xu US Risk Aversion Index (BEXRA) of Bekaert et al. (2022). A new measure of time-varying risk aversion that can eventually be calculated using observable financial data at high frequencies.	https://www.nancycxu.net/risk-aversion-index .
EPU	The global economic policy uncertainty index of Baker et al. (2016).	We obtain data of the monthly global economic policy uncertainty of Baker et al. (2016) from http://www.policyuncertainty.com/global_monthly.html . Baker et al. (2016) present two versions of the global economic policy uncertainty index, one based on current-price GDP and the other on PPP-adjusted GDP. The empirical results using these two measures are very similar, and we present the results in this study using the one based on current-price GDP measures.
PR	The global political risk relative to the US political condition. It is constructed as in Filippou et al. (2018).	Authors' calculations based on monthly country-level political risk score data from the International Country Risk Guide (ICRG) database.

This table presents the definitions and data sources of variables used in empirical analysis.

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